A Multi-Messenger Technosignature and Anomaly-Detection Campaign for Omega Centauri

Tim Swanson

The Omega Centauri Society / Post Oak Labs

tim@postoaklabs.com

October 2026

Draft v3.5 (last revised 2026-10-22) — third paper of the set; companion to *The Macro Transcension Hypothesis* (Paper A), the inward-migration review (Paper B), the migration economics (Paper D), the engineered-IMBH systems paper (Paper E), the joint accretion bound (Paper F), the residual X-ray census (Paper G), and the mass-tension paper (Paper H); prepared for omegacentauri.me

Abstract

Omega Centauri (NGC 5139), the most massive Galactic globular cluster and the probable stripped nucleus of an accreted dwarf galaxy, presents a unique conjunction of observational circumstances: the strongest current candidate for an intermediate-mass black hole (IMBH) in the Galaxy, anchored by seven stars moving above the local escape velocity (Häberle et al., 2024b); a formally unresolved factor-of-several tension between kinematic lower bounds ($\geq$ 8,200 $M_\odot$) and a pulsar-timing upper bound ($\lesssim$ 6,000 $M_\odot$; Bañares-Hernández et al. 2025); complete electromagnetic silence to the deepest radio and infrared limits yet placed on this cluster, and among the deepest placed on any globular-cluster core (Mahida et al., 2026; Chen et al., 2026); and a southern declination optimal for the newest southern-hemisphere facilities. No dedicated technosignature search of ω Cen has ever been reported at any wavelength. We present a coordinated, hypothesis-agnostic, multi-messenger campaign of eight instrument-matched programs (enumerated in Section 1) spanning infrared imaging, radio timing and SETI, astrometry, gravitational waves, neutrinos, gamma rays, the optical time domain, and archival channels, addressing conventional astrophysics (IMBH reality, mass, and spin; cluster dynamics) and technosignature hypotheses with the same data. For each program we state quantitative sensitivities, time requests, decision thresholds, and explicit falsification criteria, including negative results: with the acceleration estimator calibrated to the proper-motion precision the oMEGACat catalogue itself reports, direct astrometric acceleration detection of the fast stars stands at 0.21σ today and stays below 1σ at nominal parameters before ~2040, so the decision-grade astrometry routes through photocentric-wander and reference-frame measurements instead. Total cost is $\lesssim$ US$10M over 2026–2040, of which the cash lines are dominated by archival analysis and commensal observing, though the facility request includes ~650 h of dedicated MeerKAT time, 65 h of JWST and 50 h of CTAO. A per-mille mass and spin measurement is available from LISA if an inspiral is caught in band, at a target-specific probability of $\sim 3 \times 10^{-7}$ over a four-year mission; the probable LISA deliverable is instead a resolved-source count for the cluster's remnant population. Every null result constrains conventional astrophysics, and no anomaly claim advances without confirmation from at least two independent messengers; under realistic outcomes the campaign adjudicates the astrophysical hypotheses, while the technosignature hypothesis is constrained only along specific low-probability branches.

Keywords: Omega Centauri; NGC 5139; intermediate-mass black holes; technosignatures; SETI; multi-messenger astronomy; millisecond pulsars; gravitational waves; neutrino astronomy

Contents

1 Introduction **5**
- 1.1 The case for Omega Centauri now . . . 5
- 1.2 The technosignature rationale . . . 6
- 1.3 Design principles . . . 7
- 1.4 Structure of this paper . . . 7

2 The target and the constraint landscape **7**
- 2.1 Adopted parameters . . . 7
- 2.2 The mass tension . . . 8
- 2.3 Three hypotheses, fixed in advance . . . 8

3 Program 1: JWST infrared accretion and waste-heat limits **10**
- 3.1 Science case . . . 10
- 3.2 Current and scheduled JWST programs . . . 11
- 3.3 Implementation . . . 11
- 3.4 Decision criteria . . . 13

4 Program 2: radio deep imaging, narrowband SETI, and pulsar timing **13**
- 4.1 Science case . . . 14
- 4.2 Implementation: MeerKAT era (2026–2030) . . . 14
- 4.3 Implementation: SKA era (2029–) . . . 18
- 4.4 Decision criteria . . . 18
  - 4.4.1 Confusion per band . . . 18

5 Program 3: astrometry with HST, Gaia DR4, Roman, and ELT/MICADO **19**
- 5.1 Sensitivity of the direct acceleration test . . . 19
- 5.2 Three astrometric measurements . . . 20
- 5.3 Decision criteria . . . 23

6 Program 4: The LISA forecast **23**
- 6.1 Science case . . . 24
- 6.2 Occurrence rates . . . 24
- 6.3 The spin measurement as the H_2 test . . . 25
- 6.4 Deliverables before launch . . . 26

7 Program 5: KM3NeT/ARCA neutrino monitoring **26**
- 7.1 Science case and geometry . . . 26
- 7.2 Implementation . . . 26
- 7.3 Decision criteria . . . 28

8 Program 6: archival Fermi-LAT and CTAO-South gamma rays **28**
- 8.1 Science case . . . 28
- 8.2 Implementation . . . 28
- 8.3 Decision criteria . . . 29

9 Program 7: optical/IR time domain with Rubin/LSST and archival anomalies **29**
9.1 Science case . . . 29
9.2 Implementation . . . 29
9.3 Decision criteria . . . 30

10 Program 8: Supplementary archival channels **30**

11 Coordination, joint statistics, and data policy **31**
11.1 The two-messenger rule, operationalized . . . 31
11.2 Alert latency and standing authorizations . . . 32
11.3 Data policy . . . 32
11.4 Program summary . . . 34

12 Cost and timeline **34**
12.1 Costing . . . 34
12.2 Timeline . . . 35

13 The decision structure through 2040 **36**
13.1 Amendment C-A1 (2026-08-26, pre-data) . . . 36
13.2 Amendment C-A2 (2026-09-03, pre-data) . . . 39
13.3 Amendment C-A3 (2026-10-20, pre-data) . . . 40
13.4 Amendment C-A4 (2026-10-22, pre-data) . . . 40

14 Significance conventions across the companion papers **41**

15 Conclusion **41**

16 Revision record **42**

1 Introduction

1.1 The case for Omega Centauri now

Three independent developments between 2024 and 2026 have transformed Omega Centauri (ω Centauri, NGC 5139) from one interesting globular cluster among many into arguably the single best-posed multi-messenger target in the Galaxy outside the Galactic Centre.

First, the IMBH candidacy became concrete: Häberle et al. (2024b), using a proper-motion catalogue of 1.4 million stars built from over 500 HST epochs (Häberle et al., 2024a), identified seven stars within 3 arcsec of the cluster centre moving faster than the local escape velocity (five robust; two carried as a sensitivity row): stars that require a bound central mass of $\geq$ 8,200 $M_\odot$ from their velocities alone, and $\geq$ 21,100 $M_\odot$ at 99 per cent confidence, acceleration-model-informed, when acceleration limits are included.[1] Independent N-body modelling finds a best-fitting in-situ-grown IMBH of ~ 4.7–5.1×10^4 $M_\odot$ (González Prieto et al., 2025). Against this, a joint analysis of stellar kinematics and millisecond-pulsar (MSP) timing places a 3σ upper limit of $\sim 6 \times 10^3$ $M_\odot$ on any central point mass, favouring instead an extended ≈ 2–3×10^5 $M_\odot$ component of stellar remnants (Bañares-Hernández et al., 2025). The constraints are formally inconsistent under their stated assumptions. A factor-of-several disagreement between two mature measurement techniques, centred on the nearest IMBH candidate in the sky, warrants a focused observational campaign to resolve it.

Second, the electromagnetic silence became quantitative: the most sensitive radio observation yet made of ω Cen (~170 h with ATCA, reaching 1.1 μJy beam^{-1} rms) detects nothing at any proposed cluster centre, bounding a putative IMBH's accretion at a model-specific fraction $\lesssim 4 \times 10^{-3}$ of the Bondi rate under Mahida et al.'s own radiative/jet assumptions (Mahida et al., 2026); that fraction is not the accretion efficiency $L_{\rm bol}/(\dot{M}_{\rm B} c^2)$ (see Paper F's convention table, Swanson 2026c), extending two decades of radio non-detections of globular-cluster IMBHs (Strader et al., 2012; Tremou et al., 2018). JWST NIRCam and MIRI photometry of the central field likewise finds no source with the spectral energy distribution of an accreting IMBH (Chen et al., 2026). Whatever sits at the centre of ω Cen is dark to a depth that is a scientific result.

Third, the instrument landscape shifted decisively in the campaign's favour within an eighteen-month window centred on this writing: the Vera C. Rubin Observatory began science observations in late 2025 (the Pre-LSST period), with the ten-year Legacy Survey of Space and Time (LSST) formally begun, as announced on 2026 June 30 (Ivezić et al., 2019; Vera C. Rubin Observatory, 2026); the Nancy Grace Roman Space Telescope completed construction, with launch announced for 2026 August 30 (launch dates can move; we quote the announced date once and do not repeat the hedge) (WFIRST Astrometry Working Group et al., 2019); Gaia Data Release 4, with a 5.5-year astrometric baseline, is confirmed for 2026 December 2 (Gaia Collaboration et al., 2016); KM3NeT/ARCA reached 51 of 230 planned detection units (as of early 2026) with real-time alerts entering commissioning (Adrián-Martínez et al., 2016; The KM3NeT Collaboration, 2025); the first telescopes of CTAO-South are, as of early 2026, arriving at Paranal (Cherenkov Telescope Array Consortium et al., 2019); SKA-Mid is assembling toward science verification from 2027 and shared-risk operations from 2029 (Braun et al., 2019); ELT first light is scheduled for 2029 (Davies et al., 2021); and LISA, formally adopted by ESA in January 2024, is in hardware development for a 2035 launch (Colpi et al., 2024). Nearly every facility on this list is

[1] The companion mass-tension analysis (Paper H) builds only on the firm 8,200 $M_\odot$ velocity-only bound from the five robust stars; it does not carry the 21,100 $M_\odot$ acceleration-informed figure, which depends on an acceleration model rather than kinematics alone. The same fence applies here: the 21,100 $M_\odot$ figure appears in Table 1 and on the mass axis of Figure 1 for context and enters no branch of the decision tree of Figure 6, all of whose fences are stated against the velocity-only bound or against masses measured later in the campaign.

southern-hemisphere or all-sky; ω Cen, at declination $-47.5°$, sits in the sweet spot of all of them. A campaign organized now can ride this entire wave at marginal cost.

1.2 The technosignature rationale

This paper is the observational companion to a speculative hypothesis paper (Swanson, 2026b, hereafter Paper A), which argues that rapidly spinning massive black holes in dense old stellar systems are thermodynamic attractors for hypothetical computation-optimizing civilizations, and that ω Cen is the most accessible system satisfying the resulting selection criteria. We neither assume nor argue for that hypothesis here. The campaign is designed to be *hypothesis-agnostic* in the specific sense that every observation in it is independently justified by conventional astrophysics: IMBH demographics (Greene et al., 2020), cluster dynamics, pulsar timing, accretion physics at the lowest Eddington ratios, and EMRI astrophysics (Amaro-Seoane, 2018). The technosignature dimension adds analysis channels (narrowband drift searches, burst-coincidence triggers, anomaly statistics) to data that would be taken anyway, in the cost-effective tradition of commensal SETI (Tarter, 2001; Wright et al., 2022; Lacki and DiKerby, 2025), consistent with the archival anomaly-detection strategy recommended by the Dyson Minds workshop for black-hole-hosted intelligence (Curtis et al., 2026), and aligned with recent calls for integrated, multi-channel technosignature-search strategies as the efficient way to constrain Fermi-paradox hypotheses (Crawford, 2026).

Three considerations nevertheless justify making the technosignature dimension explicit rather than incidental. First, globular clusters are almost entirely unexplored SETI territory, and the documented record contains no search of ω Cen at any wavelength. The earliest globular-cluster technosignature search was a passive one: Cohen et al. (1980) re-analysed existing hydroxyl and water maser-survey spectra of 25 clusters, taken for astronomical reasons, to bound beacons in a sample of up to 10^7 stars, and ω Cen was absent from all three component surveys; the method, mining spectra taken for other purposes, is the archival re-use on which this campaign's commensal programs rest. The only dedicated optical SETI of globular clusters, coincidence observations of 30 southern clusters with repeat visits, recorded one laser-like event toward 47 Tuc in December 2008 that further searches of the same region did not see again, and its target list is not published, so whether ω Cen was among the thirty is unknown (Bhathal, 2010). The first dedicated globular-cluster survey at radio wavelengths, a FAST pilot of five northern clusters, was published only this year and could not observe ω Cen from FAST's latitude (Huang et al., 2026b); Gaia-based dynamical metrics for prioritizing clusters as technosignature targets have followed (Huang et al., 2026a). ω Cen itself is listed in the Breakthrough Listen Exotica Catalog, entry 344 of the Prototype sample, typed as a stripped nucleus; its observation status is not public (Lacki et al., 2021). Second, ω Cen is a privileged target on entirely generic arguments: 10^7 stars of age 12.08 Gyr (0.75 Gyr spread; systematics dominate the mean; Clontz et al. 2024) in a beam a few arcminutes across, a possible IMBH, and (as the probable nucleus of an accreted dwarf; Hilker and Richtler 2000; Ibata et al. 2019) an independent galactic chemical-evolution history, so that any ETI prior that weights stellar age and number density at all concentrates sharply here. Third, the high-energy technosignature channels (neutrino bursts from hypothetical black-hole-based computation; Dvali and Osmanov 2023) happen to be testable at ω Cen essentially for free, because the cluster's declination makes it an up-going source for the Mediterranean neutrino telescopes. Where Paper A's specific predictions sharpen a test, we say so explicitly and label the dependence; readers uninterested in that hypothesis may strike those sentences without loss to the campaign.

Epistemic status: The observational programs of Sections 3–10 stand on conventional astrophysics and are stated in the empirical register. The technosignature interpretation of any anomaly they return belongs to the speculative register, is conditional on the optimization premise of Papers A and B, and is adjudicated by the quantitative framework of Paper E rather than by this campaign.

1.3 Design principles

Five principles govern the campaign design, and we state them up front because they discipline everything that follows.

(i) Dual-use data. Every observing program must produce publishable conventional astrophysics on its own; the technosignature analysis is a parallel pipeline on the same photons (or neutrinos, or strain data), never a dedicated expenditure that a null result would waste.

(ii) Honest sensitivity accounting. Where a measurement cannot work at nominal parameters, we say so quantitatively. Section 5 demonstrates this standard: the direct acceleration test on the fast stars, superficially the most natural follow-up to Häberle et al. (2024b), sits at 0.21σ on the current catalogue and stays below 1σ before ∼2040 for plausible masses once the estimator is tied to that catalogue's own reported precision, and we restructure the astrometric program accordingly.

(iii) Two-messenger adjudication. No anomaly claim survives on one channel. Every positive trigger (a neutrino multiplet, a narrowband line, an infrared transient) must specify in advance which independent channel confirms or kills it, with combined significance assessed by pre-registered methods (Section 11).

(iv) Pre-registered thresholds. Detection and falsification thresholds are stated numerically before the data arrive (Tables 6 and 5), following standard pre-registration practice.

(v) Parasitism on funded science. The decisive measurements (IMBH reality, mass, and spin) will be made by LISA as part of its core science program whether or not anyone organizes a campaign (Colpi et al., 2024; Babak et al., 2017). The campaign's job is to ensure that when those numbers arrive, the contextual data (timing, astrometry, electromagnetic limits, high-energy monitoring) already exist to interpret them.

1.4 Structure of this paper

Section 2 reviews the target and the present constraint landscape. Sections 3–10 present the eight programs, each with science case, technical implementation, sensitivity estimates, and decision criteria: (1) JWST infrared accretion and waste-heat limits; (2) deep radio imaging, narrowband SETI, and millisecond-pulsar timing with MeerKAT, transitioning to SKA-Mid; (3) astrometric monitoring with HST, Gaia DR4, the Nancy Grace Roman Space Telescope, and ELT/MICADO; (4) a LISA extreme-mass-ratio-inspiral forecast; (5) a KM3NeT/ARCA neutrino burst pipeline exploiting ω Cen's up-going geometry from the Mediterranean; (6) Fermi-LAT archival analysis and CTAO-South follow-up; (7) optical/infrared time-domain and archival anomaly searches with Rubin/LSST and oMEGACat; and (8) supplementary archival channels (LIGO–Virgo–KAGRA continuous waves; Pierre Auger ultra-high-energy cosmic rays). Section 11 describes cross-program coordination, alert protocols, and joint statistics. Section 12 presents costing and timeline. Section 13 assembles the decision tree through 2040. Section 15 concludes.

2 The target and the constraint landscape

2.1 Adopted parameters

Table 1 collects the parameters adopted throughout. We use the oMEGACat kinematic distance of 5.49 ± 0.06 kpc (Häberle et al., 2025), consistent within systematics with the Gaia EDR3 parallax distance

Table 1: Adopted parameters for ω Centauri (NGC 5139).

Quantity	Value	Source
Distance	5.49 ± 0.06 kpc	Häberle et al. (2025)
Cluster mass	$\approx 4 \times 10^6\, M_\odot$	Harris (1996); Baumgardt and Vasiliev (2021)
Stellar count	$\sim 10^7$	Harris (1996)
Mean stellar age	12.08 Gyr (0.75 Gyr spread; systematics dominate)	Clontz et al. (2024)
Origin	stripped dwarf-galaxy nucleus	Hilker and Richtler (2000); Ibata et al. (2019); Souza et al. (2026)
Kinematic centre (J2000)	RA $13^{\rm h}26^{\rm m}47.24^{\rm s}$, Dec $-47°28'46.5''$ ($\pm 1''$)	Anderson and van der Marel (2010)
Known millisecond pulsars	19	Dai et al. (2020); Chen et al. (2023); Colom i Bernadich et al. (2026)
Central γ-ray source	4FGL J1326.7−4729 (MSP ensemble)	Dai et al. (2020, 2023)
Angular scale	$1'' = 0.0266$ pc	—

(Soltis et al., 2021) and the combined-catalogue value (Baumgardt and Vasiliev, 2021). At this distance $1''$ subtends 0.0266 pc, so the entire fast-star region ($r < 3''$) fits within 0.08 pc, and the cluster core ($r_c \approx 2.4'$) within ∼4 pc.

2.2 The mass tension

Table 2 and Figure 1 summarize the central-mass constraints. Two features matter for campaign design. First, the disagreement is not marginal: the velocity-only kinematic lower bound (8,200 $M_\odot$) exceeds the pulsar-timing point-mass upper bound (6,000 $M_\odot$, 3σ) outright. At least one analysis is wrong, or incomplete, in an instructive way: for instance through the radial distribution assumed for the MSPs, the foreground/membership treatment of the fast stars, or the possibility that the central mass is genuinely extended (Bañares-Hernández et al., 2025; Zocchi et al., 2019; Breen and Heggie, 2013). Second, the tension is resolvable on a known schedule: more pulsars and longer timing baselines tighten the upper bound as roughly $1/\sqrt{N_{\rm psr}\, T^2}$ (Section 4); Roman and Gaia DR4 sharpen the astrometric frame within two years (Section 5); and LISA, if a compact-object inspiral is caught, measures the mass to ~ 0.1 per cent (Babak et al., 2017).

See Paper F App. A for the shared-parameter table across the set, cross-referencing this campaign's distance and core-radius conventions against the other papers' own choices.

2.3 Three hypotheses, fixed in advance

Following Paper A we carry three competing hypotheses through every program:

H_0 **(gas-starved IMBH or near-IMBH).** A genuine central point mass in the $\sim 10^4\, M_\odot$ range exists and is silent for the ordinary reason: a relaxed, gas-poor, 12-Gyr-old cluster supplies essentially nothing to accrete. Recent survival analyses find that IMBHs formed early in dense clusters are plausibly retained to the present day in ω Cen-like hosts, supporting the prior on this branch (Martinez et al., 2026). Default and most parsimonious if the kinematic bounds hold.

Table 2: Current constraints on the central dark mass of ω Cen (cf. Paper A's mass-constraint table, Table 6 as of v2.8; retrace by content, not table number, if Paper A's table order changes again).

Constraint	Value	Method / source
Lower bound (velocities only)	$\geq 8{,}200\,M_\odot$	5 robust fast stars (7 monitored), HST proper motions (Häberle et al., 2024b)
Lower bound (with accelerations, acceleration-model-informed)	$\geq 21{,}100\,M_\odot$ (99%)	same stars, acceleration limits (Häberle et al., 2024b)
N-body growth models	~ 4.7–$5.1 \times 10^4\,M_\odot$	cluster-evolution simulations (González Prieto et al., 2025)
Upper bound (point mass)	$< 6 \times 10^3\,M_\odot$ (3σ)	stellar kinematics + MSP timing (Bañares-Hernández et al., 2025)
Favoured alternative	extended 2–$3 \times 10^5\,M_\odot$	same analysis (Bañares-Hernández et al., 2025)
Timing (joint MeerKAT–Parkes)	insensitive at 10^3–$10^4\,M_\odot$; $< 10^5\,M_\odot$ (90%)	MSP timing (Colom i Bernadich et al., 2026)
Confirmed stellar-mass BH	$4.46^{+1.22}_{-1.01}\,M_\odot$ (94-yr binary, $a = 31$ AU)	HST/JWST astrometry (Whitaker et al., 2026)
Historical upper limits	$\lesssim 10^4\,M_\odot$ (model-dep.)	HST proper motions (van der Marel and Anderson, 2010); see also Noyola et al. (2008); Zocchi et al. (2019)

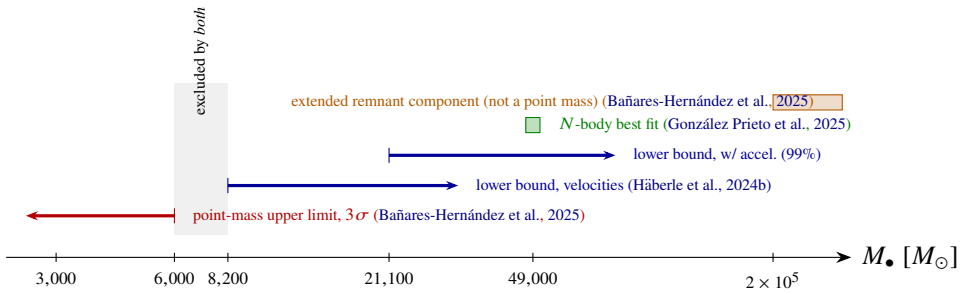


Figure 1: The central-mass constraint landscape (logarithmic mass axis). The pulsar-timing point-mass upper limit and the fast-star kinematic lower bounds leave no allowed point mass: at least one constraint must yield. The exclusions are not symmetric: the timing limit applies to a *point mass*, while the fast-star bounds apply to *any* bound central mass interior to the fast-star orbits, so an extended remnant component evades the former but not the latter. The campaign is structured to determine which constraint yields, on a known schedule, using instruments that share no dominant systematics.

H_1 **(extended remnant component).** No IMBH; the central mass is a spatially extended cluster of stellar remnants (Bañares-Hernández et al., 2025; Breen and Heggie, 2013; Zocchi et al., 2019). Default and most parsimonious if the timing bound holds.

H_2 **(anomalous/managed system).** A central IMBH whose observational properties are shaped by technology, per Paper A. Carries a low prior; becomes interesting only through conjunctions that H_0 and H_1 jointly fail to explain (e.g. near-extremal spin on a gas-starved hole; repeated coincident multi-messenger bursts; statistically anomalous depletion of the loosely bound core population).

The campaign's purpose is to measure the system well enough that H_0 and H_1 are decided on their merits, with H_2's distinctive residues either appearing in the data or being retired by them, rather than to hunt for H_2. These labels map one-to-one onto the hypothesis set of the companion adjudication framework (Swanson, 2026a, hereafter Paper E): H_0 corresponds to Paper E's H_q (quiescent astrophysical IMBH), H_1 to $H_{\rm sub}$ (sub-IMBH remnant alternative), and H_2 to $H_{\rm eng}$ (engineered system).

3 Program 1: JWST infrared accretion and waste-heat limits

Bottom line. Deep two-epoch JWST imaging of the core is the decisive observable: an accretion-like point source, or a mid-infrared excess over the stellar-population SED carrying the warm/cool swarm colour discriminant. The current best limits are archival: radiative output $\lesssim 10^{-9}$ of Eddington, radio accretion $\lesssim 4\times10^{-3}$ of the Bondi rate, and X-ray $L_X \lesssim 1.6\times10^{30}$ erg s^{-1}. The target sensitivity is ~5–10 nJy (3σ) at F444W, a bolometric limit of order 10^{30} erg s^{-1}, and a tightened ceiling of $P_{\rm comp} \lesssim 10^3$–$10^4\,L_\odot$ on managed throughput. The result feeds the deepest multi-band quiescence statement available before LISA, and any excess or variability enters the coordination protocol of Section 11.

3.1 Science case

For any central mass $M_\bullet$, the Eddington luminosity is $L_{\rm Edd} \simeq 1.26\times10^{31}\,(M_\bullet/M_\odot)$ W: 1.0×10^{35} W at 8,200 $M_\odot$, 2.5×10^{35} W at $2\times10^4\,M_\odot$, 6.2×10^{35} W at $4.9\times10^4\,M_\odot$. The existing non-detections already establish extreme quiescence: the radio limit bounds accretion at $\lesssim 4\times10^{-3}$ of the Bondi rate (Mahida et al., 2026), the archival Chandra limit bounds the X-ray output at $L_X \lesssim 1.6\times10^{30}$ erg s^{-1} (Haggard et al., 2013), and the JWST photometry bounds the radiative output at $\lesssim 10^{-9}$ of Eddington (Chen et al., 2026), among the most extreme quiescence levels measured for any black hole. The published JWST analysis, however, is an archival by-product rather than an optimized deep program, and the gain a purpose-designed campaign buys depends on which quantity is being compared. In flux the deep program improves on Chen et al.'s own archival depths by 22× at F770W and 39× at F1000W before crowding, and by 7.4× and 13× after the adopted penalty. In Eddington ratio the gain is far larger, from the $\lesssim 10^{-9}$ the archival photometry supports to 4×10^{-14}–4×10^{-13} at $2\times10^4\,M_\odot$, a factor 2.5×10^3 to 2.5×10^4, because the bolometric conversion is not the same at the two depths. The two figures are quoted separately here, since "an order of magnitude" matches neither. The archival $\lesssim 10^{-9}$ anchor is itself a bolometric reading of Chen et al. (2026): their F444W depth converts to $\sim4\times10^{-12}$ of Eddington monochromatically, and the looser published figure carries their SED assumption rather than ours. The deep program adds the two dimensions the archival data lack: variability (accretion at these rates is expected to flicker; a managed or artificial suppression of accretion, the H_2 residue, is not) and waste heat (a mid-infrared excess over the stellar population model is the classic Dyson-type technosignature; Dyson 1960; Wright et al. 2014; Hsiao et al. 2021, and at ω Cen the relevant solid angle is a few arcseconds, not a galaxy).

3.2 Current and scheduled JWST programs

Two approved JWST programs already execute pieces of this campaign, and the deliverables below are stated as increments over them. GO 5137 (PI Seth), "Weighing the Intermediate Mass Black Hole in Omega Centauri," obtains a deep NIRSpec IFU mosaic of the central field: line-of-sight (LOS) velocities for ∼70 stars in the innermost region, including all four fast movers inside 1″, together with the deepest search to date for accretion-emission signatures at the candidate position (Seth et al., 2024). GO 8322 extends the astrometric time base with dedicated JWST monitoring of the fast stars (JWST GO 8322 team, 2026). That changes three things for Program 1: (i) the accretion-emission search is no longer virgin territory — the deep-imaging deliverable is the order-of-magnitude *photometric* depth gain, the second epoch for variability, and the 10–25.5 μm waste-heat axis (the long-wavelength limit of MIRI imaging), none of which the IFU program provides; (ii) the contingent spectroscopic environment test (below) becomes an extension of GO 5137's footprint rather than a first observation; and (iii) the LOS velocities feed directly into the astrometric program, where they convert two-dimensional proper motions into three-dimensional velocity vectors and open a radial-velocity acceleration channel (Section 5.1).

3.3 Implementation

Deep imaging (new observations). NIRCam F200W/F356W/F444W at 20 ks per filter plus MIRI F770W/F1000W at 15 ks per filter, in two epochs separated by ∼6 months: ∼65 h total. In the crowded core the limits are confusion- rather than photon-limited. The NIRCam figures are ETC-class point-source depths scaled to the requested 20 ks exposure and degraded by an assumed ∼2.9× crowding penalty, reaching ∼5–10 nJy (3σ) across F200W/F356W/F444W; the empirical-PSF-subtraction gain, anchored to the oMEGACat astrometric catalogue (Häberle et al., 2024a; Nitschai et al., 2023), is named as an assumption to be verified by scene simulation, not a measured recovery. Chen et al.'s own archival NIRCam photometry of this field reaches 660 nJy–1.6 μJy at their 515 s exposure (Chen et al., 2026), which implies a crowding penalty of ∼29–70× without PSF subtraction and puts the adopted ∼2.9× assumption in view as an assumption rather than a demonstrated anchor. At MIRI the ETC-class point-source depths at 15 ks are ≈70 nJy (3σ) at F770W and ≈170 nJy at F1000W before crowding, degrading to ≈200 and ≈500 nJy respectively after the crowding penalty discussed next. At the ω Cen core stellar density ($\rho_0 \simeq 3 \times 10^3\ M_\odot\ \mathrm{pc}^{-3}$; $\sim 10^7$ stars in the field) the NIRCam F444W pixel scale (0.063″) resolves of order fifty stars per resolution element in the inner arcsecond, and the 90 per cent injection depth is a property of the local surface density and the PSF-fitting algorithm rather than the instrument. Table 3 breaks the scalar ∼3 crowding factor into per-filter completeness versus photon limits, using the same Chen et al. 90 per cent completeness curves that set the Paper F infrared leg (their Table 1, scaled by exposure time; Paper F Appendix 14 and `miri_sensitivity.json`, ETC 6.0). For each filter the ETC photon-limited 3σ depth at the requested exposure and the 90% injection-recovery depth are compared; the adopted depth is the shallower. At the requested 20 ks (NIRCam) and 15 ks (MIRI) the completeness depth is shallower in every filter, so the program is crowding-limited and its exposure is sized to that wall (Table 3). Doubling the integration improves the photon limit by $1/\sqrt{2}$ but moves the adopted limit by <0.1 dex, while halving it would make sensitivity bind. This pre-empts the TAC question of why 15 ks rather than 8 ks at MIRI: at 8 ks the photon and completeness depths are comparable, and at 15 ks the photon depth sits a factor ∼1.5 below the completeness wall, so the time request buys the wall rather than empty depth. The MIRI figures carry the larger systematic: RGB-tip giants are mJy-level sources at 10 μm whose extended PSF halos overlap in the core, in-flight long-wave throughput degradation must still be budgeted, and the quoted depths assume the NIRCam-calibrated ∼3× penalty transfers to the mid-IR PSF—assumptions a full scene simulation must verify before the time request is finalized.

Table 3: Per-filter depth at the requested exposure: photon limit versus 90% completeness (Chen et al. 2025, as used in Paper F) and which binds. MIRI ETC from `miri_sensitivity.json` (ETC 6.0, 10 ksec scaled as $t^{-1/2}$; rescaled 61 nJy at F770W and 115 nJy at F1000W 3σ, printed as ≈70 and ≈170 nJy before crowding); NIRCam ETC/completeness from the oMEGACat F444W injection that recovers 90% at the quoted depth, scaled by exposure and pixel scale with F356W interpolated. Adopted is the shallower; all five filters are crowding-limited at the requested depth, so exposure is sized to the completeness wall.

Filter	$\lambda_{\rm piv}$ (μm)	Exp. (ks)	ETC photon 3σ (nJy)	90% completeness (nJy)	Binds	Adopted 3σ (nJy)
F200W	2.00	20	4	12	crowding	12
F356W	3.56	20	3	9	crowding	9
F444W	4.44	20	3	8	crowding	10[a]
F770W	7.70	15	61 (70[b])	184 (200[b])	crowding	200
F1000W	10.0	15	115 (170[b])	345 (500[b])	crowding	500

[a] Printed as ~5–10 nJy (3σ) across F200W/F356W/F444W, anchored to the injection; adopted 10 nJy is the shallower end, consistent with 3× penalty. [b] Parentheses are the values quoted in the pre-table sentence; they exceed the rescaled figures by 14% at F770W and 48% at F1000W, which is a conservative round-up rather than a rounding. The adopted column takes the shallower quoted value in each case; first numbers are the rescaled ETC/completeness from `fC_calc7_miri.json`.

At 5.49 kpc, 10 nJy at F444W corresponds to a monochromatic $\nu L_\nu \approx 2.5 \times 10^{28}$ erg s^{-1}; for plausible quiescent-accretion SEDs the bolometric limit is of order 10^{29}–10^{30} erg s^{-1}, i.e. Eddington ratios of 4×10^{-14}–4×10^{-13} at $2 \times 10^4\ M_\odot$, probing a regime in which even Sgr A* would be conspicuous (Event Horizon Telescope Collaboration, 2022).

Two-epoch variability. Differential photometry between epochs is robust to the static crowding systematics that dominate the absolute limits, and the trigger is a significance on the epoch difference rather than an amplitude alone. Two epochs at per-epoch 1σ noise σ_1 give a difference noise $\sqrt{2}\,\sigma_1$, so an amplitude f on a source of flux F is detected at $fF/(\sqrt{2}\sigma_1)$. At the adopted NIRCam depth (10 nJy at 3σ, so $\sigma_1 = 3.3$ nJy and $\sqrt{2}\sigma_1 = 4.7$ nJy) a 3σ difference at 20 per cent amplitude needs $F > 71$ nJy, and 47 nJy at 30 per cent; at the post-crowding F1000W depth (500 nJy at 3σ) the corresponding floors are 3.5 μJy and 2.4 μJy. Draft v2.6's "≥20–30 per cent for any source above ~30 nJy at NIRCam (above ~0.5 μJy at F1000W)" is withdrawn: those fluxes give 1.3σ and 0.42σ at 20 per cent amplitude, so they would promote noise. The pre-registered trigger is a 5σ difference, which at 20 per cent amplitude requires $F > 118$ nJy at NIRCam and $F > 5.9\,\mu$Jy at F1000W, and the floors degrade further by the per-filter crowding penalty of Table 3, which the F1000W figure already carries and the NIRCam figure does not. Any source clearing that threshold flags an accretion candidate and triggers the coordination protocol of Section 11. Two-epoch photometry should not be mistaken for a test of Paper E's regulation statistic R, the variability *deficit* in the MAD flux-eruption band (10^{-2}–10^{-1} Hz for a $2 \times 10^4\ M_\odot$ hole; cycle times of 10–100 s), which requires fast-cadence monitoring. At the fiducial mass that is an X-ray measurement, not a radio one: a compact self-absorbed synchrotron source low-pass filters the eruption band at the source, and interstellar scintillation adds variability uncorrelated with accretion on top, so the radio band carries no discriminating power for R; the channel activates at $L_X \gtrsim 10^{34}$–10^{35} erg s^{-1}, i.e. in flare or TDE states (Swanson, 2026a). Accordingly, if accretion is ever detected at that level in any band, a fast-cadence X-ray contingency activates (pointed timing plus the archival X-ray channel of Program 8), targeted at the eruption band, with Program-2 radio monitoring serving as context rather than as an R measurement.

Waste-heat analysis (archival + new). The 10–25.5 μm field photometry is fitted against the synthetic stellar-population SED constructed from the oMEGACat spectroscopic catalogue (Nitschai et al.,

2023). The primary waste-heat deliverable is a *point-source* SED excess with a colour discriminant: the configurations surviving Paper E's feasibility analysis are cool swarms at 10^2–10^3 AU, which subtend 0.018″–0.18″ at 5.49 kpc and are unresolved inside the F1000W PSF core (0.33″), so morphology carries no information inside that envelope and the discriminant is spectral: a 50–300 K near-blackbody with no silicate features, against the AGB dust shells that dominate the false-positive population (Swanson, 2026a). Paper E's swarm SED forward model (Appendix C as of v2.1, "Priors, likelihoods, and computation"; retrace by content, not appendix letter, if Paper E's appendix order changes again), a two-temperature swarm SED integrated against the nine MIRI imaging filters, supplies this discriminant quantitatively: any point-source excess is tested against the predicted warm/cool flux ratio across F560W–F2550W rather than a single-band flux, which any AGB dust shell of comparable brightness will not reproduce over the same filter set. We retain the extended statistic (a spatially coherent excess in the central 0.1 pc, i.e. 3.8″, exceeding 3σ over the population model) but relabel it: it is a hypothesis-agnostic anomaly channel, no longer the test of the surviving waste-heat configurations. The deliverable is quantitative through Paper E's waste-heat floor, $L_{\rm waste} \geq 10^{-4}\, P_{\rm comp}$: the current JWST-derived limit converts to a ceiling on sustained computational power of $P_{\rm comp} \lesssim 10^3$–$10^4\, L_\odot$, and the ceiling tightens linearly with mid-infrared depth (Swanson, 2026a). One temperature qualifier bounds every MIRI-derived ceiling: a 50 K swarm peaks near 60 μm and is essentially invisible at ≤ 25.5 μm, so the ceiling applies to the warm ($\gtrsim 150$ K) end of the configuration space only, and the cold end is unconstrained by any instrument in this campaign. The same fit yields conventional science: the most precise mid-IR census of the post-main-sequence population in any globular cluster.

Spectroscopic environment test (Cycle 6+, contingent). The approved GO 5137 IFU observations (Section 3.2) already cover the innermost arcseconds; the contingent extension proposed here is a wider NIRSpec IFU mosaic of the inner 0.5 pc (∼125 h), activated only if astrometry or timing strengthens the IMBH case, mapping abundance anomalies (Li enhancement, refractory depletion) that would discriminate between ordinary tidal-disruption debris chemistry (Hills, 1988; Gezari, 2021) and the engineered-feeding residue of H_2. We do not request this time until the trigger condition is met.

3.4 Decision criteria

Null result (expected under H_0, H_1, and mature-H_2 alike): bolometric limit $\lesssim 10^{30}$ erg s^{-1} recorded (radiative output $\lesssim 10^{-12}$ of Eddington across the contested mass range), complementing the radio Bondi-efficiency bound and jointly constraining any radiatively inefficient flow via the fundamental plane (Merloni et al., 2003; Mahida et al., 2026). A static detection consistent with a quiescent accretion SED supports H_0 and immediately sharpens every dynamical program (the source position becomes the kinematic centre). A variable detection triggers multi-messenger follow-up. A $> 3\sigma$ mid-IR excess (point-source with the swarm colour discriminant, or extended with no stellar counterpart) is the single strongest electromagnetic anomaly the campaign can produce and would be adjudicated under Section 11 rules.

4 Program 2: radio deep imaging, narrowband SETI, and pulsar timing

Bottom line. The observable that bears most directly on the central mass is the radial profile of millisecond-pulsar accelerations, which separates a central point mass from an extended remnant component only if per-pulsar acceleration noise is at or below 10 (km s^{-1})^2 pc^{-1} (Section 4); the narrowband SETI limit and the deeper continuum image ride on the same sessions. The current best limits are the 3σ point-mass cap of $\sim 6 \times 10^3\, M_\odot$ from the joint kinematic-timing analysis and the latest joint MeerKAT–Parkes

insensitivity at 10^3–$10^4\ M_\odot$ with a 90-per-cent ceiling of $10^5\ M_\odot$. The target is ten well-timed pulsars on ten-year baselines, pushing the bound to the 3,000–4,000 $M_\odot$ level or finding the profile, plus a first ω Cen technosignature limit near 4×10^{15} W EIRP over the full cluster. This program feeds decision point D2 directly.

4.1 Science case

The radio program carries three loads at once. (i) *Continuum*: a deeper interferometric limit (or first detection) on the central compact source, extending Mahida et al. (2026) and the globular-cluster IMBH radio campaigns (Strader et al., 2012; Tremou et al., 2018). (ii) *Narrowband SETI*: to our knowledge the first dedicated technosignature search of any kind at ω Cen (the one candidate for precedence, Bhathal's thirty-cluster optical survey, does not publish its target list), closing the gap left by the FAST pilot survey's latitude limit (Huang et al., 2026b). (iii) *Pulsar timing*: the binding constraint on the central mass comes from the cluster's MSP population (five discovered with Parkes, Dai et al. 2020, timed to $\dot{\nu}$ within 3.5 yr, Dai et al. 2023; then thirteen more with MeerKAT/TRAPUM, Chen et al. 2023; Stappers and Kramer 2016). The joint MeerKAT–Parkes timing analysis now extends both the census and the constraints: it reports a nineteenth discovery, PSR J1326−4728S, bringing the known population to nineteen, and its measurements are insensitive to an IMBH of 10^3–$10^4\ M_\odot$ while placing a 90 per cent upper limit of $< 10^5\ M_\odot$ on the central mass (Colom i Bernadich et al., 2026). The same analysis finds an anomalously high fraction of isolated and black-widow systems among the cluster's MSPs, which bears directly on the MSP radial-distribution priors on which the Bañares-Hernández et al. (2025) timing bound leans, since the spatial distribution of a dynamically processed population need not follow the assumed profile. The path to resolving the mass tension runs directly through more pulsars and longer baselines (Bañares-Hernández et al., 2025; Chen et al., 2025).

4.2 Implementation: MeerKAT era (2026–2030)

Timing. Bi-weekly L-band sessions (~26 per year, ~3–5 h each) on all nineteen MSPs with MeerKAT (Jonas and MeerKAT Team, 2016), producing times of arrival at several-μs precision for the brightest objects (the cluster's MSPs are 10–30 μJy sources; sub-μs timing is not on offer). Line-of-sight acceleration precision from timing scales steeply with baseline ($\sigma_a \propto T^{-5/2}$ for white noise); by analogy with the mature 47 Tuc and Terzan 5 programs (Freire et al., 2017; Prager et al., 2017), five years of MeerKAT data yield per-pulsar acceleration uncertainties of order 10^{-10} m s^{-2} for the best-timed objects on the white-noise extrapolation (the forecast below finds that unknown intrinsic spin-down, not timing noise, sets the effective noise). We do not extrapolate to the 10^{-11} decade: the 47 Tuc and Terzan 5 precedents rest on 20–25-yr baselines, and DM variation and red timing noise break the white-noise $T^{-5/2}$ scaling before that precision is reached (van Haasteren and Levin, 2013). For scale, a $4 \times 10^4\ M_\odot$ point mass imposes $a \approx 6 \times 10^{-8}$ m s^{-2} at $r = 0.3$ pc: the discriminating signal is not the detection of acceleration (already achieved; Dai et al. 2023; Bañares-Hernández et al. 2025) but the radial profile of accelerations and jerks across the MSP population, which distinguishes a point mass ($a \propto r^{-2}$) from an extended remnant component. Commensal search observations are expected to add 5–10 MSPs (the luminosity function is far from exhausted; Chen et al. 2023), and the population-level point-mass upper bound tightens approximately as $1/\sqrt{N_{\rm eff}}\,(T_0/T)^{\geq 1}$; this is an ensemble, bound-level scaling, distinct from the per-pulsar white-noise acceleration scaling $\sigma_a \propto T^{-5/2}$ quoted above, because the ensemble constraint is limited by the pulsars' unknown line-of-sight positions and intrinsic spin-down as well as by timing noise. The effective sample size is not 19. The released timing record (`paper/h/data/pulsars.json`) carries published timing solutions for twelve of the nineteen pulsars: seven (A, B, C, D, E, H, K) on the full

Murriyang UWL campaign, 2020 April 1 to 2025 May 20, a 5.13-yr baseline, and five (G, I, L, N, Q) on MeerKAT-only solutions spanning 2021–2025, taken as 4.0 yr. Six others (F, J, M, O, P, R) have no timing solution in the record and one more (S) is a single-epoch discovery. The per-pulsar acceleration sensitivities, $\sigma_{a,i} \propto P_i\,\sigma_{t,i}/T_i^2$, differ through the spin periods (2.5–19 ms), so the inverse-quadrature sum with weights $w_i = 1/\sigma_{a,i}$ gives $N_{\rm eff} = (\sum_i w_i)^2/\sum_i w_i^2 = 10.0$ on that baseline assignment, replacing the 9.8 carried through Draft v2.6, which stated no baselines and was therefore not reproducible. Two other readings of the same twelve bracket it: weighting by baseline alone gives 11.4, and equal weights give 12. The timing residual enters $\sigma_{a,i}$ as a common factor (several-μs ToA precision, not sub-μs) and cancels in $N_{\rm eff}$, which is a ratio of weight moments; rescaling every $\sigma_{t,i}$ by a common constant leaves $N_{\rm eff}$ unchanged to machine precision, as `paper/figs/fC_r9_checks.py` verifies. The spin period does not cancel and is not claimed to. The ensemble sensitivity is dominated by the seven pulsars on the longer baseline, the shortest-period of which (H, 2.52 ms) alone carries 17 per cent of the total weight. Ten well-timed pulsars with ten-year baselines push a 6,000 $M_\odot$ bound to the 3,000–4,000 $M_\odot$ level if no point-mass signal emerges, or find one. That forecast and Colom i Bernadich et al. (2026)'s statement that the present measurements are insensitive to 10^3–10^4 $M_\odot$ are consistent rather than contradictory: the insensitivity is a statement about the current 4–5-yr baselines, and the projection buys its factor from doubling the baseline against the $T^{-5/2}$ per-pulsar scaling and from the ensemble term, with the reservations on red noise and DM variation above. The projection is a forecast on this paper's own account and is not attributed to that analysis.

Forecast requirement (pre-registered). The profile discriminator just described is the campaign's primary near-term test, and it is at present asserted rather than demonstrated: most of the MSPs sit well outside the central 0.1 pc, and their unknown line-of-sight positions degrade the radial information the test needs. Before the D2 data lock (decision point D2 of the D1–D4 sequence defined in Section 13) we will therefore complete and publish a mock-population forecast: point-mass and extended-component potentials injected at the real MSP sky positions, line-of-sight positions marginalized over the cluster density profile, and realistic timing noise applied, reporting the recoverable Bayes factor (or $\Delta\chi^2$) as a function of baseline and census size. An order-of-magnitude expectation frames the exercise: with 19–25 MSPs, 5–10-yr baselines, and per-pulsar acceleration uncertainties of order 10^{-10} m s^{-2} (the white-noise extrapolation; see the forecast below), the profile difference between a 2×10^4 $M_\odot$ point mass and a 2–3 $\times 10^5$ $M_\odot$ extended component is of order 10^{-9}–10^{-8} m s^{-2} for the innermost pulsars at their favourable, near-plane-of-sky assumed 3D positions. The pre-registered success criterion is a median discrimination of $\Delta\chi^2 \geq 9$ between the two named potentials at the D2 epoch; if the forecast falls short, the D2 decision weight shifts to the astrometric channels of Section 5.

Forecast as executed. The forecast has been run (`paper/figs/fC_calc7b_d2forecast_v2.py`; 200 realizations per cell, three injected truths × five noise floors × four census sizes). The forecast returns two statistics, and they behave differently enough that quoting one without the other misrepresents the test. Both are signed positive toward the extended potential throughout. The *fixed-parameter* statistic compares the two named potentials with no free parameters, and it is the one D2 routes on: under the injected 2×10^4 $M_\odot$ point mass its median runs from −0.27 to −23 across the noise ladder at the current census of nineteen, and it routes to the point-mass branch in 99.5, 84 and 9.5 per cent of realizations at the three lowest floors and in none at the two highest, where the realizations are almost all inconclusive ($|\Delta\chi^2| < 9$); under the injected 2.5×10^5 $M_\odot$ extended component the medians run +0.20 to +436 and it routes to the extended branch in 100, 93, 30, 1 and 0 per cent of realizations across the same five floors. The *free-parameter* statistic, which maximises each family over its own mass (and, for the extended family, scale) grid, cannot be used for routing at all: under the point-mass truth its median is +0.003 to +0.02 across the noise ladder, which carries no routing information, and it recovers the correct sign in only 21–48

per cent of realizations. The reason is structural rather than a defect of implementation. The extended family reduces to the compact one as its scale radius goes to zero, so maximising over the larger family can only improve the fit; the statistic measures fit improvement and carries no information about which model generated the data. The order-of-magnitude "$\Delta\chi^2$ of order tens ($\ln K \sim 5$–15)" expectation carried through Draft v2.6 corresponds to nothing the forecast returns for either statistic, and is withdrawn in favour of the numbers above.

The pre-registered criterion is met at 3.09 and 10 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$ and missed for both named truths at 31.6 and above. The formal timing error is of order 10^{-13} $\mathrm{m\,s^{-2}}$ for the pulsars with timing solutions, so the noise that matters is the unknown intrinsic spin-down, which enters each measured $\dot{P}$ alongside the cluster acceleration (Colom i Bernadich et al., 2026). We take 31.6 to 64 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$ as the D2-epoch noise. The fallback is therefore in force: D2 weight lies with the astrometric channels of Section 5 unless an independent constraint on intrinsic spin-down reaching 10 is demonstrated before the D2 data lock.

The equivalence $\Delta\chi^2 = 2\ln K$ holds only when the two models being compared have the same dimension, and it is retracted here for the free-parameter case: the extended family carries one extra parameter, and at the current census the free-parameter $\Delta\chi^2$ and $2\ln K$ disagree in sign at the timing-limited floor (+0.02 against −0.14) and at the noisiest (+0.01 against −0.02). It remains exact for the fixed-parameter statistic, which is why the routing thresholds are stated on that one.

Where the test has no authority. Both named potentials differ in enclosed mass within the pulsar region by a factor of 12.5, so a statistic distinguishing them is partly reading mass rather than profile shape. A third truth isolates this: a Plummer component of $3.3 \times 10^4\,M_\odot$ at $a = 2$ pc, chosen to enclose the same mass as the $2 \times 10^4\,M_\odot$ point mass at the median projected pulsar radius (3.13 pc). The routing statistic does not recover it at any floor: the median is negative at all five (from −21.5 at 3.09 to −0.30 at 200 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$), and it routes the genuinely extended component to the point-mass branch in 98.5, 80 and 7.5 per cent of realizations at the three lowest floors; at the two highest floors every realization is inconclusive. D2's authority is therefore limited to discriminating the two potentials it names. An extended component whose enclosed mass resembles the point mass is misrouted at the two lowest floors and inconclusive at 31.6 and above, and that limitation is carried into the routing rule of Amendment C-A1 rather than left implicit. A complementary dynamical diagnostic is available from the same data: an IMBH measurably alters the degree of energy equipartition of the host cluster (Aros and Vesperini, 2023), so the equipartition profile inferred from the kinematic catalogues provides an independent axis along which the point-mass and extended hypotheses separate.

Narrowband SETI. Commensally with every timing session, a 1-Hz-resolution spectrometer backend records the full primary beam (FWHM $\sim 1^\circ$ across most of L band, containing the whole cluster). Processing follows standard drift-search practice (Enriquez et al., 2017): Doppler drifts of ± 4 $\mathrm{Hz\,s^{-1}}$ at L band (a fractional convention, $\dot{\nu}/\nu \approx 3 \times 10^{-9}$ $\mathrm{s^{-1}}$; the absolute window scales with observing frequency for any S-band search), on–off cadence against reference pointings, and a candidate-advance threshold of SNR > 15. The sensitivity limits below are quoted at the conventional SNR = 10 of Eq. (1); the two numbers serve different roles (limit-setting versus pipeline promotion) and are stated separately to keep them from being conflated. Seventy hours over two years reach a minimum detectable EIRP of $\sim 4 \times 10^{15}$ W at 5.49 kpc. Deeper is available on the same commensal data at no extra facility cost, since the limit improves as $t^{-1/2}$: the full five-year timing allocation accumulates 390–650 h (26 sessions per year of 3–5 h) and lowers the threshold by a factor 2.4–3.0 (to ~ 1.3–1.7×10^{15} W). The 70-h figure is the two-year milestone quoted for the first publication, and the five-year figure is the programme's end state. That threshold follows from the standard narrowband relation

$$\mathrm{EIRP}_{\mathrm{min}} \;=\; 4\pi d^2\,\mathrm{SNR}\,\mathrm{SEFD}\,\sqrt{\delta\nu/t}, \tag{1}$$

with $d = 5.49$ kpc, SNR = 10, $\delta\nu = 1$ Hz, and $t = 70$ h. The SEFD entering Eq. (1) is an array figure, not

an antenna figure: the ≈ 430 Jy value is the per-antenna SEFD of a single 13.5-m MeerKAT dish (Jonas and MeerKAT Team, 2016), and an $N \approx 60$-antenna sum improves on it by $\sqrt{N}$ to ≈ 55 Jy incoherently, or by N to ≈ 7 Jy in a coherent tied-array beam. The quoted 4×10^{15} W is the incoherent, full-cluster figure; a coherent tied-array pointing reaches $\approx 5 \times 10^{14}$ W but covers only an arcseconds-scale beam. These limits are quoted within the half-light radius: the EIRP limit degrades by 2× at the half-power radius of the primary beam, a transmitter at the ∼0.8–1° tidal radius ($r_c\ 10^c \approx 0.81°$ from Harris 1996; a Nature-paper King-profile fit gives 0.81°, Häberle et al. 2024b) sits at or beyond the half-power point at the top of L band, and the S-band primary beam ($\sim 40'$) does not contain the cluster at all. The incoherent full-cluster limit is below the FAST pilot's $\sim 10^{16}$ W thresholds at its closer, northern targets (Huang et al., 2026b); it is the first limit of any kind for this cluster, and it is sufficient to detect any transmitter exceeding $\sim 2 \times 10^2$ times the EIRP of the Arecibo planetary radar ($\sim 2 \times 10^{13}$ W; Enriquez et al. 2017, a modest output for an energy-rich civilization) from any of $\sim 10^7$ stars in the beam simultaneously. For comparison with FAST's per-target accounting, a single $5''$ coherent follow-up beam at the core contains $\sim 10^3$ stars, so even the deep coherent limit is a per-thousand-star figure rather than a per-star one.

Backend requirement: spatial RFI discrimination. An incoherent full-primary-beam sum has no spatial discrimination: every signal within $\sim 1°$ enters the data product, and there is no on/off inside a single pointing. The dominant contaminant for a 2026+ campaign is not fixed-frequency terrestrial RFI but LEO mega-constellation downlinks sweeping through L band with Doppler drifts inside the searched $\pm 4\ \mathrm{Hz\,s^{-1}}$ window; candidate rates of 10^4–$10^5\ \mathrm{hr^{-1}}$ are routine in current data. The mitigation is architectural rather than procedural: simultaneous coherent beams with candidate cross-rejection, the BLUSE architecture of Czech et al. (2021), in which a candidate appearing in widely separated beams is rejected as local. Tiling the half-light region with $5''$ coherent beams would require thousands of beams; the realistic design forms a modest set of anti-coincidence beams plus a reduced-baseline subarray whose wider coherent beams cover the core at intermediate SEFD. We state this as a hard backend requirement, because without it the false-candidate budget, not the radiometer equation, sets the program's labour cost.

Scintillation-aware confirmation. The 5.49-kpc, DM ≈ 100 $\mathrm{pc\,cm^{-3}}$ sightline scintillates strongly at L band: diffractive scintillation modulates a narrowband signal by factors of several on minutes-to-hours timescales, the case treated for SETI by Cordes et al. (1997). A criterion requiring re-detection in a single independent session would therefore reject a genuine steady transmitter with order-unity probability whenever the confirming epoch lands in a scintillation null. The pre-registered confirmation policy is scintillation-aware: a candidate surviving the cross-rejection screen is scheduled for multiple re-observation epochs spaced by at least the refractive timescale, and confirmation significance is computed under exponential-intensity statistics rather than a constant-flux assumption, so that a non-detection in any single epoch demotes rather than kills the candidate.

Continuum. The summed continuum visibilities from the timing campaign provide a ∼100+ h synthesis dataset at L/S band, and the depth budget is set by confusion, not thermal noise: at the $\sim 6''$ resolution of a naturally weighted L-band image the confusion limit is formalism-dependent: the classical one-source-per-beam criterion gives ≈ 0.35 μJy, while the Condon (1974) self-consistent criterion (cutoff at five times the rms confusion; Condon 2007) gives ≈ 3.5 μJy, against a thermal noise of 0.41 μJy in 100 h (Table 4). The confusion-versus-thermal ordering at $6''$ therefore depends on the formalism, while at UHF confusion dominates in both; 19+ steep-spectrum MSPs sit at the position of interest either way. The useful deliverables are therefore long-baseline, uniform-weighted imaging of the central arcminute (where the confusion limit is far deeper), UHF/S-band spectral decomposition separating the steep-spectrum pulsar population from any flat-spectrum accretion candidate, and in-beam variability of compact sources; combined with the ATCA 7.25-GHz limit (Mahida et al., 2026) these constrain both the flat-spectrum (jet) and steep-spectrum (pulsar) interpretations of any future central source candidate.

4.3 Implementation: SKA era (2029–)

SKA-Mid science verification is expected from 2027, shared-risk operations from 2029, and early full operational cycles in the early 2030s (Braun et al., 2019). The ω Cen program transfers wholesale: timing precision improves by the sensitivity ratio (factor ∼4–5 over MeerKAT for these declinations), the MSP census plausibly doubles, and the narrowband EIRP thresholds drop by the same factor, toward $\sim 10^{15}$ W incoherent and $\sim 10^{14}$ W coherent, on a southern target FAST cannot see. (The ngVLA, the other next-decade radio flagship, is not an option here: at $\delta = -47.5°$ ω Cen is inaccessible from its planned sites.) By LISA launch, the pulsar acceleration map will be the best non-GW constraint on the central potential in any globular cluster.

4.4 Decision criteria

The timing program is the campaign's primary near-term discriminator between H_0 and H_1 only where the noise condition of the forecast is met. At the noise set by unknown intrinsic spin-down it returns mostly inconclusive verdicts, and D2 weight then shifts to the astrometric channels. Where it is met, a smooth r^{-2} acceleration profile centred on the kinematic centre to within $\max(1'', r_\bullet(M_\bullet))$, the wander-inflated tolerance of Section 5.2, at $\geq 10^4\,M_\odot$ retires H_1; a persistently tightening upper bound below the kinematic lower bounds forces revision of the fast-star analysis (membership, foreground, or binarity systematics) and shifts the campaign's centre of gravity to H_1. For SETI: any re-detected narrowband candidate enters the two-messenger protocol; a null is published as the first ω Cen technosignature limit. Paper A dependence (labelled): under H_2, P3 of that paper predicts no leakage radiation, so SETI nulls carry no evidential weight against H_2; they constrain only conventional beacon scenarios. Consistent with this, Paper E's Bayes-factor analysis finds that additional radio depth beyond current limits moves $\ln K$ only weakly for every hypothesis pairing (Swanson, 2026a); the case for the deeper continuum limit therefore rests on the H_0/H_1 accretion physics, not on technosignature discrimination.

4.4.1 Confusion per band

Classical confusion estimates differ by about an order of magnitude at these resolutions depending on formalism, so Table 4 carries the two defensible readings per band rather than one number: the one-source-per-beam cutoff at half its flux, and the Condon self-consistent criterion, whose cutoff sits at five times the rms confusion (Condon, 2007); the framework is the flux-density error-distribution formalism of Condon (1974). Both readings use the same deep 1.4-GHz count law ($N(>S) \approx 220\,(S/\mathrm{mJy})^{-1}\,\mathrm{deg}^{-2}$ across 10 μJy–1 mJy, flattening toward the Euclidean slope above 1 mJy), whose normalization carries about a factor-of-two uncertainty; the band-to-band ratios are more robust than the absolute scale. The independent cross-check supports that factor and no better: Vernstrom et al. (2014) measure an rms confusion of ∼1.2 μJy beam^{-1} at 3 GHz and 8″, where the count law and criterion adopted here give 2.8 μJy beam^{-1}. The spread is therefore a factor of ten within a band and a factor of sixty across the table (0.15 to 9.1 μJy beam^{-1}), and the absolute scale is not settled by anything in this paper. Which column binds depends on the band: at L and S the thermal noise and the classical cutoff are comparable (∼0.35 against 0.41 μJy beam^{-1}), so a Condon-limited reading would put the L-band program 8× above its thermal floor and make confusion, not integration time, the wall; at UHF the Condon reading exceeds the thermal floor by more than an order of magnitude under every normalization, which is why the deep continuum limit is quoted at L band with UHF and S band carried as spectral-index leverage. Uniform weighting at 4″ is the mitigation the program actually buys, halving both readings. An earlier version of the accompanying JSON carried a third column anchored to an NVSS value of 0.45 mJy at 45″; that figure is the NVSS rms noise rather than its confusion limit (Condon et al., 1998), the same law and

criterion give ∼0.19 mJy at that resolution, and the column is withdrawn rather than reported. Numbers in `figs/fC_calc7_confusion.json` and `figs/fC_calc7_condon_fix.json`, the latter superseding the former's anchor status.

Table 4: Confusion (μJy beam^{-1}) under two formalisms, against the 100-h thermal noise. The Condon column applies the criterion of Condon (2007): the cutoff is set at five times the rms confusion.

Band	ν / θ	1-per-beam	Condon $q = 5$	thermal
L	1.4 GHz / 6″	0.35	3.46	0.41
S	3.0 GHz / 6″	0.35	2.03	—
UHF	0.8 GHz / 8″	0.62	9.11	—
L, uniform	1.4 GHz / 4″	0.15	1.54	—

5 Program 3: astrometry with HST, Gaia DR4, Roman, and ELT/MICADO

Bottom line. The single decisive observable is the Gaia DR4 re-derivation of the fast-star velocity excess on the improved absolute frame, the cheapest measurement in the campaign that can either firm the mass tension into a contradiction or dissolve it; Roman photocentric wander and MICADO inner-star discovery carry the longer-term mass discrimination. The current best astrometry is the oMEGACat catalogue, against which direct acceleration detection stands at 0.21σ and stays below 1σ before ∼2040 (Eq. 3). The targets are the DR4 proper-motion improvement (several-fold over DR2-era ties), few-μas-class wander measurement from Roman, and new fast stars at $r < 0.02$ pc from MICADO. The DR4 re-derivation feeds decision point D1 in 2027; wander and inner stars feed D2 and the LISA prior.

5.1 Sensitivity of the direct acceleration test

The intuitively obvious astrometric test (watch the seven fast stars curve) does not work on a useful timescale, which we now demonstrate quantitatively. The proper-motion acceleration induced by a central mass $M_\bullet$ at projected radius r is

$$a = \frac{GM_\bullet}{r^2} \approx 4.4 \times 10^{-7} \left(\frac{M_\bullet}{2 \times 10^4\, M_\odot}\right) \left(\frac{r}{0.08\,\mathrm{pc}}\right)^{-2} \mathrm{m\,s^{-2}}, \tag{2}$$

which at $d = 5.49$ kpc corresponds to $\approx 5.3 \times 10^{-4}$ mas yr^{-2} (mean projection factors of order unity absorbed). Against this, one least-squares chain sets the sensitivity of every astrometric forecast in this section and in Figure 2. For N epochs spread over a baseline T at per-epoch positional precision σ_{pos}, the same fit of $x = x_0 + \mu t + \frac{1}{2} a t^2$ returns

$$\sigma_\mu = \frac{\sqrt{12}\,\sigma_{\mathrm{pos}}}{\sqrt{N}\,T}, \qquad \sigma_a = \frac{26.83\,\sigma_{\mathrm{pos}}}{\sqrt{N}\,T^2}, \qquad \text{whence} \qquad \sigma_a = \frac{7.75\,\sigma_\mu}{T}, \tag{3}$$

where the third form eliminates σ_{pos} and N between the first two. That third form is what we use for existing data, because it needs no assumed epoch count: it is fixed by the proper-motion precision the catalogue itself reports, and any epoch distribution sparser than uniform degrades the curvature term faster than the linear one, so it bounds what a real cadence can deliver rather than assuming a cadence.

The oMEGACat catalogue reports $\sigma_\mu \approx 6.6\,\mu$as yr^{-1} within $r < 1.5'$, the region containing the fast stars, and $\approx 11\,\mu$as yr^{-1} across the survey at $m_{\mathrm{F625W}} \approx 18$, on a 20-year baseline (Häberle et al., 2024a).

Equation (3) turns the central-region value into $\sigma_a = 2.6 \times 10^{-3}$ mas yr^{-2}, a 0.21σ measurement of the nominal signal and 0.51σ of the heavy one; the survey-median value gives 0.12σ. Extending HST monitoring to 2028 (a 26-year baseline) reaches 0.40σ on the nominal signal under $T^{-2.5}$ scaling, and the nominal signal crosses 1σ at $T \approx 38$ yr, which is the origin of the ~2040 date quoted in the abstract and in Section 1. These figures supersede the $\sigma_a \approx 1 \times 10^{-3}$ mas yr^{-2}, 0.5σ and $\sim 0.7\sigma$ values carried through Draft v2.6, which rested on an inferred per-epoch $\sigma_{\rm pos} = 0.02$ mas. Equation (3) cannot reconcile that inference with the catalogue's own σ_μ at any epoch count: at $N = 2T$ it implies $\sigma_\mu = 0.55\,\mu$as yr^{-1}, twelve times better than the catalogue reports, while reproducing 1×10^{-3} mas yr^{-2} at $\sigma_{\rm pos} = 0.02$ mas instead requires $N \approx 1.8$ epochs. The recomputation is in `paper/figs/fC_r9_checks.py`.

For facilities that do not yet exist the calibration is unavailable and the forecast has to state its cadence. A four-year ELT/MICADO campaign at ~50–100 μas per epoch (Davies et al., 2021) at 2 epochs yr^{-1} ($N = 2T$) reaches $\sigma_a \approx (3.0\text{–}5.9) \times 10^{-2}$ mas yr^{-2} (4.2×10^{-2} at the fiducial 70 μas), equivalent to $\sigma_\mu \approx 21\,\mu$as yr^{-1} under Eq. (3); this is worse than the HST archive because acceleration precision scales as $T^{-2.5}$ and four years is short (Figure 2). Direct 5σ curvature detection on the known fast stars at the calibrated 20-year sensitivity would require $M_\bullet \gtrsim 5 \times 10^5\,M_\odot$ at $r = 0.08$ pc, or a star at $r < 0.016$ pc at the nominal mass, or baselines of ~50 yr (heavy mass) to ~70 yr (nominal); ELT-class astrometry at 2 epochs yr^{-1} reaches 5σ on the heavy mass at ~30 yr and 38–44 yr on the nominal mass at 50–70 μas. We therefore retain the fast-star monitoring (cheap, and it protects against the lucky cases, since González Prieto et al. 2025-mass holes and undiscovered inner stars are both live possibilities) and route the decision-grade astrometry through the three astrometric measurements below.

The approved GO 5137 spectroscopy (Section 3.2) adds two levers this budget did not include. First, the LOS velocities of the fast movers convert their two-dimensional proper motions into three-dimensional velocity vectors; since the LOS component can only increase a star's total speed, each measurement can only tighten the escape-velocity lower bound of Häberle et al. (2024b), star by star. Second, repeat NIRSpec visits open a radial-velocity-drift acceleration channel independent of the proper-motion curvature budget above. The expected LOS acceleration at the fiducial parameters of Eq. (2) is $a_{\rm los} \approx 4.4 \times 10^{-7}$ m s^{-2} ≈ 0.014 km s^{-1} yr^{-1}. Assuming per-epoch RV precision of ~3 km s^{-1} (crowded-field NIRSpec on faint members) and annual visits over five years with white noise, the recoverable drift uncertainty is $\sigma_{\dot v} \approx \sigma_{\rm RV}\sqrt{12}/(\sqrt{n}\,T) \approx 0.9$ km s^{-1} yr^{-1}: a factor ~70 above the nominal signal, the same verdict as the proper-motion channel (Figure 2, lower panel). The channel is nonetheless worth running because it shares none of the proper-motion systematics (frame tie, geometric distortion) and inherits the full r^{-2} gain for any inner star: at $r = 0.01$ pc the drift signal reaches ~0.9 km s^{-1} yr^{-1}, reaching parity ($\sim 1\sigma$) with $\sigma_{\dot v}$ within a five-year GO 5137 extension; a 3σ detection requires a ~decade baseline or higher-cadence visits.

5.2 Three astrometric measurements

(i) Photocentric wander (Roman). An IMBH of mass $M_\bullet$ in dynamical equilibrium with a stellar bath executes Brownian motion with velocity $\sigma_\bullet \sim \sigma_*\sqrt{m_{\rm eff}/M_\bullet}$ relative to the cluster centre of mass. The effective mass is the mass-weighted mean $m_{\rm eff} = \langle m^2\rangle/\langle m\rangle$, not the number-weighted $\bar m$, because the velocity diffusion scales with $n\langle m^2\rangle$ while the drag scales with $n\langle m\rangle$ (Chatterjee et al., 2002; Merritt et al., 2007). The distinction is worth a factor of a few here and it runs in the program's favour: for the segregated mass function adopted in Paper E, $m_{\rm eff} \simeq 2.3\,M_\odot$ against $\langle m\rangle = 0.54\,M_\odot$, so the amplitude is ~2.7× larger than an equal-mass estimate at $\bar m = 0.3\,M_\odot$ gives, ~13 μas yr^{-1} rather than ~5 at 8,200 $M_\odot$. The lighter the hole, the larger its wander: the predicted amplitude differs by ~17 per cent between 6,000 and 8,200 $M_\odot$ (the square root of the mass ratio), and by a factor ~2.4 between 8,200 and $4.9 \times 10^4\,M_\odot$. Those ratios are independent of $m_{\rm eff}$, so the pre-registered deliverable below is unaffected by the correction; only the detectability margin moves. Resolving it requires measuring the reflex motion of the innermost stellar

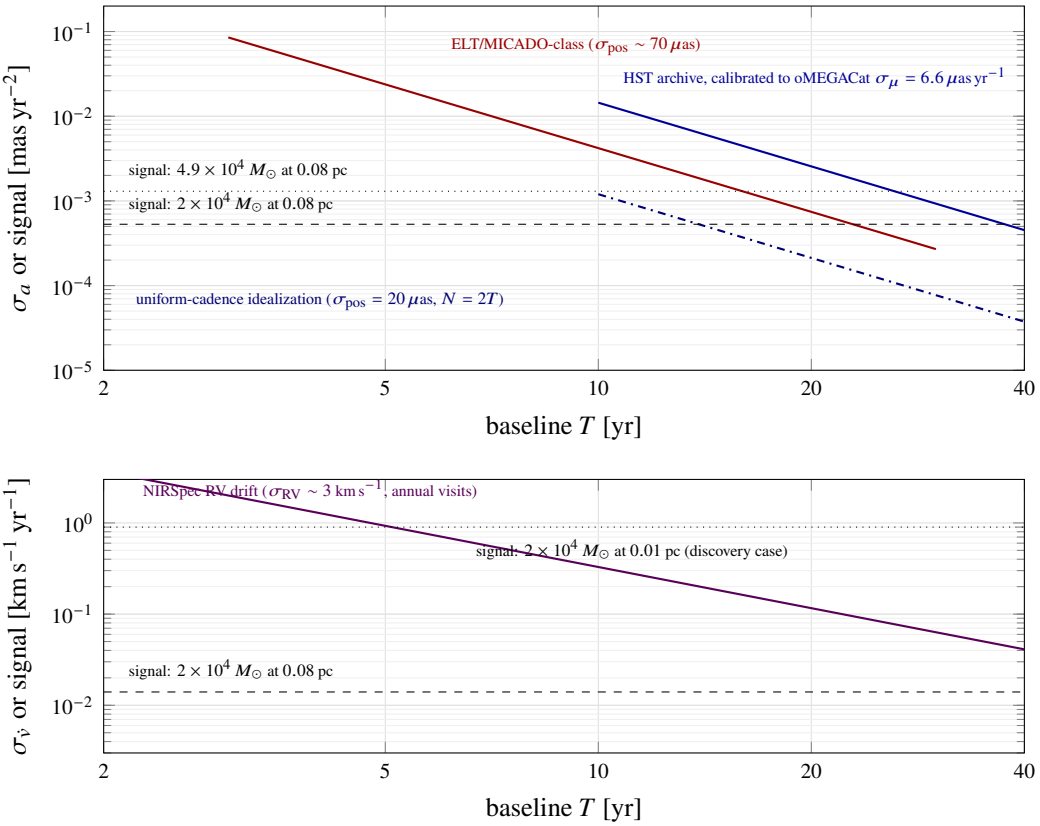


Figure 2: Why direct acceleration detection on the known fast stars is marginal. *Upper panel:* 1σ acceleration uncertainty versus baseline, against the expected signals (horizontal lines, Eq. 2). All three curves are the one least-squares chain of Eq. (3). The solid blue curve is the HST archive calibrated through $\sigma_a = 7.75\,\sigma_\mu/T$ to the proper-motion precision oMEGACat itself reports inside $r < 1.5'$ ($\sigma_\mu = 6.6\,\mu$as yr^{-1} at $T = 20$ yr, scaled as $T^{-2.5}$); the survey-median $\sigma_\mu = 11\,\mu$as yr^{-1} lies a factor 1.67 above it and is not drawn. The dash-dotted curve is the uniform-cadence idealization plotted in Draft v2.6 ($\sigma_{pos} = 20\,\mu$as with $N = 2T$), retained here as the bracket it is: Eq. (3) makes it equivalent to $\sigma_\mu = 0.55\,\mu$as yr^{-1}, twelve times better than the catalogue delivers, and its apparent 5σ crossings at ∼18 yr (heavy mass) and ∼26 yr (nominal) are properties of that assumed cadence rather than of the data. The calibrated curve crosses 5σ at ∼50 yr on the heavy González Prieto et al. (2025) mass and ∼70 yr on the nominal mass, both beyond the plotted range, and crosses 1σ on the nominal mass at ∼38 yr. The red ELT/MICADO curve is a forward forecast at $\sigma_{pos} = 70\,\mu$as and $N = 2T$, with no catalogue to calibrate against; it crosses 5σ on the heavy mass at ∼30 yr and does not reach 5σ on the nominal mass within the plotted baseline. *Lower panel:* the radial-velocity-drift channel opened by GO 5137-class NIRSpec monitoring (Section 5.1): at the nominal fast-star radius the verdict matches the proper-motion channel, while an inner star at 0.01 pc crosses the sensitivity curve within a few years. The astrometric program is therefore built around wander, reference-frame, and discovery measurements instead (Section 5.2).

distribution at the few-μas yr^{-1} level against the bulk cluster frame: the regime of Roman's wide-field astrometry, which delivers ~10 μas-class differential astrometry over fields vastly larger than HST's, tying the inner arcseconds to thousands of cluster reference stars at once (WFIRST Astrometry Working Group et al., 2019). Two saturation caveats bound the reference-star pool: ω Cen's brightest giants saturate Roman's detectors even in the shortest exposure modes, and the faint end is photon-starved in crowding, so we take the usable reference-star window to be roughly F146 $\approx$ 16–21 mag (a stated assumption to be verified against the as-flown detector performance; the $\gtrsim 10^3$ stars-per-field figure below already assumes it). With launch announced for 2026 August 30, a five-year cadenced guest-observer program (ω Cen lies outside the core community surveys, so epochs must be requested, but the per-epoch cost is small) delivers, as its headline result, the factor-2.4 discrimination between the light-IMBH and heavy-IMBH wander regimes, independently of individual-star orbits. The 17 per cent split inside the light regime is not a deliverable at any plausible reference-frame precision, and Draft v2.6's claim that it would support "3–5σ discrimination" is withdrawn here. The obstacle is degrees of freedom rather than photon noise. Wander is a variance, so what a campaign measures is the rms of a stochastic amplitude, and the relative error of a variance estimate on $N_{\rm dof}$ independent samples is $\sqrt{2/N_{\rm dof}}$. A 17 per cent amplitude ratio is a 37 per cent variance contrast, so resolving it needs $N_{\rm dof} > 2n_\sigma^2/0.367^2$, which is 134 independent samples at 3σ and 372 at 5σ. A five-year cadenced program delivering $\gtrsim$10 epochs measures the wander variance to about 45 per cent, four times the contrast it would have to resolve, and averaging over $\gtrsim 10^3$ reference stars sharpens the frame tie at each epoch without adding independent samples of the hole's own orbit. The factor-2.4 light-versus-heavy separation asks for $N_{\rm dof} > 1$ at 3σ and > 2 at 5σ on the same arithmetic, which $\gtrsim$10 epochs meets, and it is the only wander deliverable we pre-register.

That deliverable now carries a numeric pre-condition rather than a promise to quantify one later. The light and heavy predictions are separated by $13.5 - 5.5 = 8.0\,\mu$as yr^{-1} at $m_{\rm eff} = 2.3\,M_\odot$, so a 3σ separation requires every error combined, N-body modelling of the stellar background included, to stay below $2.7\,\mu$as yr^{-1} ($1.6\,\mu$as yr^{-1} at 5σ). Before the pre-registration locks, the wander prediction will be pushed through the González Prieto et al. (2025) N-body realizations and the model-to-model scatter quoted as the systematic floor; if that scatter exceeds $2.7\,\mu$as yr^{-1} the wander channel carries no D2 weight, and Table 5's astrometric row states the condition explicitly alongside the gate on Roman's as-flown crowded-field performance (which has never been measured on a field like this) and the F146 reference-window assumption above.

One error budget the wander calculation implies has been missing from the search geometry. The same Brownian equilibrium that gives the hole a velocity $\sigma_\bullet$ gives it a displacement: in the cluster's harmonic core ($\rho_0 \simeq 3\times10^3\,M_\odot\,{\rm pc}^{-3}$, so $\omega = \sqrt{4\pi G\rho_0/3} = 7.35\,{\rm km\,s^{-1}\,pc^{-1}}$) the one-dimensional rms offset is $r_\bullet = \sigma_\bullet/\omega$, which at $m_{\rm eff} = 2.3\,M_\odot$ is 0.048 pc (1.80″) at 8,200 $M_\odot$, 0.056 pc (2.10″) at 6,000 $M_\odot$, and 0.020 pc (0.74″) at $4.9\times10^4\,M_\odot$; the projected two-dimensional rms is $\sqrt{2}$ larger, 2.54″, 2.97″ and 1.04″ respectively. In the light-mass hypotheses this equals or exceeds the ±1″ kinematic-centre uncertainty of Table 1 and fills the 3″ fast-star region, so a fixed centre is an assumption the light hypotheses do not license. The rest of the paper carries the consequences. The centre uncertainty used in any search is inflated to $\max(1'', r_\bullet(M_\bullet))$, which is 2–3″ in the light case. The pointing tolerance for Program 8's "at the kinematic centre" trigger and for the X-ray re-registration of Section 10 is set the same way. The quoted fast-star radii are wander-smeared at $6–8\times10^3\,M_\odot$, which loosens rather than tightens the $r < 3''$ membership. The amplitudes and the required-scatter figure are recomputed in `paper/figs/fC_r9_checks.py`.

The astrometric channel itself now carries a proof of capability: Whitaker et al. (2026) used the same HST astrometric archive (2002–2023), extended with JWST imaging, to detect the reflex wobble of a main-sequence star orbiting an unseen $4.46^{+1.22}_{-1.01}\,M_\odot$ companion, the first stellar-mass black hole confirmed in ω Cen and the first astrometric black-hole detection in any globular cluster. A dynamically

measured dark remnant in the same field, with a 94-yr period and 31-AU separation recovered from curvature over a two-decade baseline, demonstrates end-to-end that the catalogue's astrometry supports orbit-grade inference at the precision this program assumes.

(ii) The absolute reference frame (Gaia DR4, 2026 December). The dominant systematic in all current inner-field astrometry is the tie between HST's relative frame and the absolute (ICRS) frame. Gaia DR4's 5.5-year solutions for the $r > 10''$ cluster members (Gaia Collaboration et al., 2016) improve proper motions by several-fold over DR2-era ties, propagating directly into the fast-star velocity vectors — and thus into the 8,200 $M_\odot$ lower bound itself, which rests on those velocities exceeding the escape speed. This is the cheapest decisive measurement in the entire campaign: a re-derivation of the Häberle et al. (2024b) bound on the DR4 frame either firms the tension into a $> 5\sigma$ contradiction with the timing bound or dissolves it. Deliverable in 2027 from archival data alone.

(iii) Inner-star discovery (ELT/MICADO, 2029+). MICADO's ∼10 mas resolution and 39-m aperture (Davies et al., 2021) penetrate the central arcsecond at magnitudes HST cannot reach in crowding, with the realistic prize being new fast stars at $r < 0.02$ pc. A single star at 0.01 pc raises the acceleration signal of Eq. (2) by a factor of 64, converting Figure 2's verdict from "decades" to "a few epochs" — this is how the Galactic-Centre program succeeded (GRAVITY Collaboration et al., 2018), and ω Cen is its natural successor target at $10^2\times$ lower mass. With ELT first light scheduled for 2029 and commissioning occupying the first year of operations, the first MICADO astrometric epochs realistically arrive in 2030–31, at two epochs per year thereafter; first acceleration-grade results by the early 2030s.

5.3 Decision criteria

The D1 sample and centre are fixed here, before DR4 lands, because a bound re-derived on a re-chosen sample cannot adjudicate anything. The primary D1 statistic is computed on the five robust fast stars of Häberle et al. (2024b), with the seven-star set (the five plus the two carried as that paper's sensitivity row) reported alongside as a sensitivity check and not as the headline; Table 2 labels the sample each bound rests on, since the monitored set and the set the bound is computed from are not the same seven. The centre is the published kinematic centre propagated to the DR4 epoch, not re-fitted to the fast stars themselves, and the aperture is the wander-inflated $\max(1'', r_\bullet)$ of Section 5.2. DR4 re-derivation (2027): if the fast-star velocity excess survives at $> 5\sigma$ on the five-star sample, H_1 requires the MSP analysis to be systematically biased, a specific and checkable claim (radial distribution priors; Bañares-Hernández et al. 2025). Roman wander (2031): light/heavy discrimination feeds the LISA prior. MICADO inner stars (2030s): any star with measured Keplerian curvature yields $M_\bullet$ to tens of per cent, independent of statistical modelling, before LISA flies.

6 Program 4: The LISA forecast

Bottom line. The decisive observable is an extreme- or intermediate-mass-ratio inspiral caught in band, which would measure the central object's mass and spin regardless of its electromagnetic state. There is no current limit to state; the program is a forecast. The target precision is fractional mass to 10^{-3}–10^{-4} and spin to 10^{-3}. Whether it is ever delivered is governed by two probabilities that differ by an order of magnitude or more, so both are quoted. The chance of a new capture during a four-year mission at the González Prieto et al. (2025) rate is $\sim 3\times10^{-7}$; the chance that a compact object is already in band, which is what LISA would actually see, is the residence-weighted occupancy, $\sim 2\times10^{-6}$ at 2 mHz rising to $\sim 3\times10^{-5}$ at 0.8 mHz for the 31 $M_\odot$ companion. The occupancy is the decision-relevant number and is what the mass split of D4 is scored against; the spin split needs a Tier-3 measurement. A Tier-3

measurement can retire the thin-disc exploitation model ($a_\star < 0.1$, T5) in the same instant it settles the mass tension, while a high spin supports the managed-system hypothesis only jointly with silence and regulation, and the probable deliverable is the Tier-1 resolved-source count rather than an inspiral.

6.1 Science case

LISA is the campaign's final stage: the only instrument that can measure the central object's mass and spin to high precision regardless of its electromagnetic state (Colpi et al., 2024; Amaro-Seoane et al., 2017). For a compact object inspiralling into a 10^4-class IMBH, matched-filter parameter estimation delivers fractional precisions of order 10^{-3}–10^{-4} on mass and 10^{-3} on spin (Babak et al., 2017; Amaro-Seoane, 2018), a deliberately conservative floor: Babak et al. (2017) reports fractional mass errors as tight as 10^{-6} for favourable configurations, and we quote the weaker end of their range throughout. At 5.49 kpc (versus the Gpc distances over which LISA expects to detect its EMRI population), any ω Cen inspiral in band during the mission would be loud, with matched-filter SNR of order 10^4–10^5 depending on companion mass (Section 6), making occurrence probability, not sensitivity, the limiting factor.

6.2 Occurrence rates

Event-rate estimates in the LISA literature imply IMRI formation rates per cluster of order 10^{-9}–10^{-6} yr^{-1} depending on IMBH mass, binary fraction, and core density (Mandel et al., 2008; Amaro-Seoane, 2018; Fragione et al., 2018; Arca Sedda et al., 2021). A target-specific value is now available and is three decades narrower: the ω Cen models of González Prieto et al. (2025), in which the IMBH grows chiefly by capturing 30–40 $M_\odot$ black holes (mean 31 $M_\odot$), give a present-day capture rate of $(4\text{–}8) \times 10^{-8}$ yr^{-1}. We adopt that figure, and the 31 $M_\odot$ companion it describes, where a single number is needed, and retain the generic range and its lighter 1.4 $M_\odot$ companion case as the class-level comparison; the two populations are not interchangeable, and pairing one rate with the other's waveform misstates occurrence by more than an order of magnitude, as the Tier-2 calculation below shows explicitly. Over a 4–10-year mission the generic range yields an in-band inspiral probability for ω Cen of $\lesssim 10^{-5}$, and the target-specific rate gives $\sim 3 \times 10^{-7}$ over four years: possibly enhanced by the cluster's unusually dense remnant population, but not by the four to five orders of magnitude required to make it likely.

The forecast is stated in three tiers. **Tier 1 (probable, no ω Cen source):** the deliverable is a resolved-source count, not a stochastic-foreground limit. ω Cen at 5.49 kpc, localized to a $\sim$ 10-arcmin core, places any in-band binary of the remnant population among LISA's individually resolved sources rather than folded into an unresolved galactic-confusion background; the deliverable this tier owes is a predicted count under H_0 and H_1 against the observed null, a stronger and more localized test of the extended-component density profile than a foreground amplitude would be. **Tier 2 (early-inspiral capture):** a compact object caught in the early, slowly evolving portion of an inspiral, in band for years to millennia before plunge depending on companion mass. This is not a distinct source class from Tier 3; it is the same inspiral intercepted earlier, and its detectability and occurrence pull in opposite directions, and both depend on which population is doing the inspiraling. For the target-specific 31 $M_\odot$ companion the González Prieto et al. (2025) rate is licensed for: $\mathcal{M}_c = 411.9\,M_\odot$, giving $\dot{f} = (96/5)\,\pi^{8/3}(G\mathcal{M}_c/c^3)^{5/3} f^{11/3} \approx 1.65 \times 10^{-12}$ Hz s^{-1} at $f_{\rm GW} = 2$ mHz, residence $f/\dot{f} \approx 38$ yr, and a target-specific Tier-2 occupancy $(4\text{–}8) \times 10^{-8}\,\mathrm{yr}^{-1} \times 38\,\mathrm{yr} \approx 2 \times 10^{-6}$; at 0.8 mHz the residence lengthens to ≈ 440 yr and the occupancy recovers to $\approx 3 \times 10^{-5}$. Detectability is overwhelming at either frequency, SNR of order 10^6 over a 4-yr mission against LISA's instrument-only $\sqrt{S_n} \approx 3 \times 10^{-20}$ Hz$^{-1/2}$ at 2 mHz. For the lighter 1.4 $M_\odot$ companion the generic class-level rate applies, not the target-specific one: $\mathcal{M}_c = 64.3\,M_\odot$, $h \approx 3.5 \times 10^{-19}$ at 2 mHz (orbital radius ≈ 0.03 AU, $\sim 10^2\,r_g$) and 5.49 kpc, residence

$\approx$ 850 yr ($\approx 10^4$ yr at 0.8 mHz), SNR of order 10^5 over 4 yr against the instrument floor (even a 0.1 $M_\odot$ companion accumulates $\sim 7 \times 10^3$), and an occupancy $(10^{-9}$–$10^{-6}\,\mathrm{yr}^{-1}) \times (10^3$–$10^4\,\mathrm{yr}) \approx 10^{-6}$–$10^{-2}$, derived from, not independent of, the Tier-3 rate. That 1.4 $M_\odot$ range is the generic-rate case, not one González Prieto et al. (2025) licenses a target-specific rate for; quote the band with both the number and the companion mass. The instrument-only $\sqrt{S_n} \approx 3 \times 10^{-20}$ Hz$^{-1/2}$ used above omits the galactic-confusion foreground, which for a 4-yr mission (Robson et al., 2019) adds 78 per cent in power at 2 mHz ($\sqrt{S_n} \approx 4.0 \times 10^{-20}$, SNR still of order 10^5 for the 1.4 $M_\odot$ case) and dominates by a factor 3.4 at 0.8 mHz ($\sqrt{S_n} \approx 4.3 \times 10^{-19}$), so the low-frequency end of the occupancy range above is bought at a sensitivity cost the raw residence numbers do not price. If any compact object is in band LISA cannot miss it, and the orbital frequency and its drift then measure $M_\bullet$ to precision intermediate between timing and a full late inspiral; but the expectation remains that the band is empty. **Tier 3 (lucky):** a genuine inspiral; per-mille mass and spin. Even in Tier 1, a spin constraint for the class arrives statistically: LISA's detected population of comparable IMBH systems at cosmological distances calibrates the natural spin distribution against which any eventual ω Cen measurement is judged. The campaign's structure ensures that even Tier 1 is decisive when combined with the pulsar acceleration map (Section 4): a resolved-source non-detection plus a point-mass timing profile isolates H_0. **Tier 4 (H_1's own mHz signature):** a fourth outcome belongs on this tier list even though it never activates H_2: a remnant subcluster dense enough to explain H_1 contains a substantial stellar-mass black-hole population, and that population produces its own resolvable mHz binaries (SNR $\sim$ few$\times 10^4$ at 2 mHz over a 4-yr mission), a distinct and unambiguous LISA signature that a null central massive object does not otherwise predict.

6.3 The spin measurement as the H_2 test

Paper A dependence (labelled): under H_2 the residue Paper A first named is near-extremal spin ($a_\star \gtrsim 0.9$) on a hole that has demonstrably accreted nothing for Gyr (the thin-disc spin-up branch of Paper E, Section 3.4; on a MAD-mode spin-up the endpoint is $a_\star \approx 0.035$–0.07, at or below the natural-capture null and below the T5 trigger $a_\star < 0.1$, i.e. indistinguishable from the null); under H_0 natural formation channels for cluster IMBHs span low-to-moderate spin, and for ω Cen specifically the growth history modelled by González Prieto et al. (2025) ($\sim 10^3$ captures of 30–40 $M_\odot$ black holes onto a 500–5000 $M_\odot$ seed, seed-epoch mass ratio $q \simeq 0.006$–0.08, final-mass ratio $\mu/M_f \simeq 7 \times 10^{-4}$) is the isotropic minor-merger regime, which under isotropic capture ($\langle \cos\iota \rangle = 0$ for the captured objects) random-walks the spin to $a_\star \sim 0.05$–0.10 (central ≈ 0.06; Swanson 2026a) rather than toward the $\chi \simeq 0.7$ attractor of comparable-mass hierarchical growth, which the random walk alone reaches only at $q \approx 0.12$, a factor ~ 175 above ω Cen's final-mass ratio. That prediction holds only while the net-alignment fraction of the captured population stays below $\epsilon_{\rm rot} \approx 0.03$ (the random-walk and coherent terms match at 0.026–0.031; Swanson 2026a); above it a coherent, unsuppressed spin-up term takes over and the low-spin prediction degrades, a premise Swanson (2026a) states explicitly and recommends checking directly against the González Prieto et al. (2025) realizations. Whether this contrast is the full low-versus-high one at this target depends on the spin prior (the decision tree of Section 13 carries Paper A's coherent-growth prior, under which a high spin alone does not discriminate); the attractor caveat applies to the class-level LISA population, not to ω Cen. A Tier 3 spin measurement can therefore retire the thin-disc exploitation model ($a_\star < 0.1$, T5) at the same instant it completes the H_0/H_1 question, while a high spin supports H_2 only jointly with silence and regulation; this is the reason this campaign's electromagnetic and dynamical groundwork must precede LISA rather than follow it. The full decision tree, including this branch and its isotropy premise, is assembled in Section 13.

6.4 Deliverables before launch

Between now and the mid-2030s the program's work is preparatory but concrete: (i) maintain the dynamical model of the inner parsec (timing + astrometry) so that any LISA source has an immediate host context; (ii) publish the ω Cen-specific EMRI/IMRI rate forecast with the post-2026 mass constraints folded in; (iii) ensure the cluster's barycentric ephemeris and distance ($\pm$0.06 kpc; Häberle et al. 2025) are maintained at the precision LISA parameter estimation will assume.

7 Program 5: KM3NeT/ARCA neutrino monitoring

Bottom line. The decisive observable is a high-energy neutrino multiplet: at least three tracks within 1° of the cluster, $E \gtrsim 10$ TeV, inside one sliding window, significant beyond 5σ post-trials with no astrophysical counterpart. The current best is the archival ANTARES+IceCube baseline, with ARCA at 51 of 230 detection units and sensitivity growing roughly linearly through the decade. The target is full-array steady-state point-source sensitivity of a few $\times 10^{-12}$–10^{-11} TeV cm^{-2} s^{-1} plus a decade-scale burst pipeline over $\sim 3 \times 10^6$ windows. A null closes the burst-mode channel at its stated strength; any candidate triggers the two-messenger protocol of Section 11.

7.1 Science case and geometry

ω Cen at declination $-47.5°$ is an up-going source for the Mediterranean: KM3NeT/ARCA observes it through the Earth, with atmospheric muons filtered out and sub-0.2° angular resolution for track events at $\gtrsim$10 TeV (Adrián-Martínez et al., 2016). For IceCube the same source is down-going (degraded but usable above ~100 TeV; Aartsen et al. 2020); the combined ANTARES+IceCube southern-sky analysis (Albert et al., 2020) defines the archival baseline. ARCA stood at 51 of 230 detection units as of early 2026, with real-time alert distribution entering commissioning; sensitivity grows roughly linearly with instrumented volume through the decade. The partially built array has already delivered a discovery-class result: the ~220-PeV track event KM3-230213A, the most energetic neutrino yet observed (The KM3NeT Collaboration, 2025), demonstrates that ARCA detects and reconstructs the rare high-energy events on which this program's burst criterion rests. The conventional science driver is a deep point-source limit on a dense old stellar system; the technosignature driver (Paper A dependence, labelled) is the burst-mode prediction of black-hole-based computation models (Dvali and Osmanov, 2023; Lacki and DiKerby, 2025), for which a cluster like ω Cen is the natural test bed and which the existing point-source pipelines do not target: their time-integrated statistics dilute rare short multiplets.

7.2 Implementation

Steady-state search (archival + ongoing). Standard unbinned-likelihood point-source analysis (Braun et al., 2008) at the ω Cen coordinates in (i) the ANTARES+IceCube combined dataset, (ii) the growing ARCA exposure. Full-array ARCA reaches $E^2\Phi \sim$ few $\times 10^{-12}$–10^{-11} TeV cm^{-2} s^{-1} at this declination over multi-year integrations — the deepest neutrino limit ever placed on a globular cluster.

Burst pipeline (new, the campaign's contribution). A pre-registered transient search in windows of 10^2, 10^3, and 10^4 s within 1° of the cluster centre, $E \gtrsim 10$ TeV, on ARCA data plus IceCube alerts. The background expectation calibrating the criterion is an atmospheric-neutrino rate above 10 TeV within a 1° cone of 10^{-7} s^{-1} (~3 events yr^{-1} in the cone), a deliberate ~20× envelope on the ~0.15 yr^{-1} that Paper A derives for the same cone from the KM3NeT 2.0 letter of intent (Adrián-Martínez et al., 2016), as

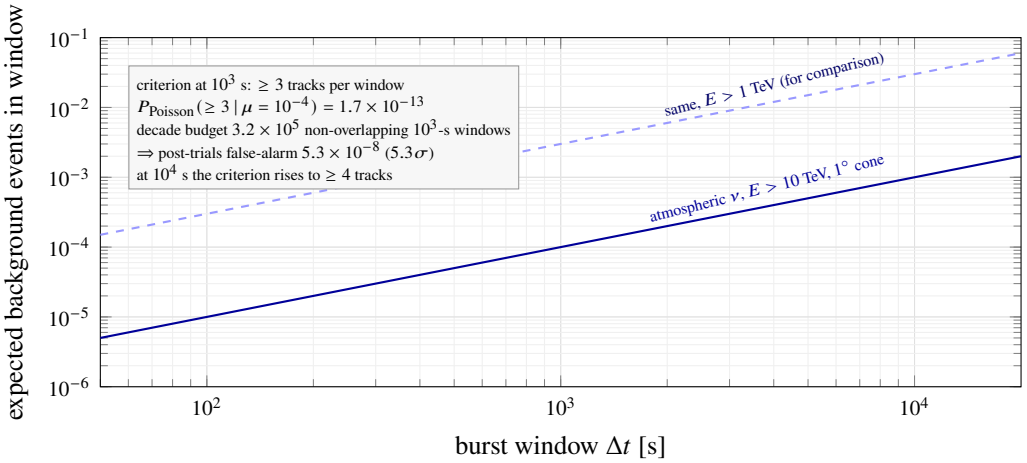


Figure 3: Design of the neutrino burst search. Expected atmospheric-neutrino background within 1° of ω Cen versus window length, for the working $E > 10$ TeV threshold (solid) and a softer 1 TeV threshold (dashed). At 10^3 s the expectation is $\sim 10^{-4}$ events, so the triplet criterion clears 5σ post-trials at 10^2 and 10^3 s under the 20× envelope; at 10^4 s it reaches only 4.4σ against that envelope, which is why the pre-registered multiplicity rises to four tracks in the longest window. Rate normalizations are order-of-magnitude, with exact values set by the detector Monte Carlo at analysis time.

inherited via Swanson 2026b, and one that likely overstates the measured up-going rate for a cone this size by one to two orders of magnitude.

The trials budget is window-dependent, and so is the criterion. A decade of monitoring holds 3.16×10^6 non-overlapping 10^2-s windows, 3.16×10^5 at 10^3 s and 3.16×10^4 at 10^4 s, so a single trials count cannot serve all three. Under the 20× envelope a triplet gives post-trials false alarms of 5.3×10^{-10} (6.1σ) at 10^2 s and 5.3×10^{-8} (5.3σ) at 10^3 s, both clearing the 5×10^{-7} action boundary, and 5.3×10^{-6} (4.4σ) at 10^4 s, which does not. The pre-registered criterion is therefore ≥ 3 tracks at 10^2 and 10^3 s and ≥ 4 tracks at 10^4 s, the last giving 1.3×10^{-9} (6.0σ); under the realistic 4.75×10^{-9} s^{-1} rate a triplet clears 5σ at all three window sizes, so the raised multiplicity costs nothing except in the envelope case it is there to cover. Draft v2.6's "$\sim 5\times10^{-7}$ ($\approx 5\sigma$)" is withdrawn: it pairs the 10^3-s expectation with the 10^2-s window count, a combination that reproduces 5.3×10^{-7} and belongs to no single window size. Windows are treated as non-overlapping for the trials count; a sliding search at finer step inflates the count by the oversampling factor and must be priced at analysis time. The recomputation is in `paper/figs/fC_r9_checks.py`. Any KM3NeT or IceCube real-time alert within 1° triggers the electromagnetic protocol of Section 11, and the alert system's first ω Cen-relevant cycle begins in late 2026.

The $\geq$ 3-track multiplet criterion reaches $p_{\text{det}} \approx 0.9$ only for burst fluences exceeding an RX-J1713-equivalent steady source by $\sim 1.1\times10^5$ in a 10^3-s window, and 2.3×10^5 for $p_{\text{det}} \approx 0.999$; the amplification scales inversely with window length, so the same p_{det} needs 10^6 at 10^2 s and 10^4 at 10^4 s (the ARCA effective-area integral, calibrated on the LOI's own event-rate table for the RX J1713 track analysis: 8.1 ν_μ tracks in 5 years after final cuts at a source declination of −39.8°, the same near-horizon visibility class as ω Cen at −47.5°). At astrophysically scaled fluences $p_{\text{det}} \ll 1$, and the excluded burst rate scales as $1/p_{\text{det}}$. The channel is therefore a test of an extraordinary hypothesis, burst fluences that no known astrophysical source produces, rather than a monitor of conventional neutrino emission.

7.3 Decision criteria

A decade-scale null at full-array sensitivity closes the burst-mode channel of Dvali and Osmanov (2023) for this system at its predicted strength — a publishable falsification regardless of H_2 (the underlying micro-black-hole physics is itself contested; Álvarez-Domínguez et al. 2024; Loeb 2024). One confirmed multiplet with no counterpart, repeated at the same location, would be among the most significant anomalies in high-energy astrophysics independent of any ETI interpretation; the protocol deliberately treats it as an astrophysical discovery first.

8 Program 6: archival Fermi-LAT and CTAO-South gamma rays

Bottom line. The decisive observable is a Fermi-LAT transient coincident with a Program-5 neutrino window, with a spectral excess above the few-GeV MSP cutoff and a CTAO TeV detection or limit as the conventional-science counterpart. The current best is the H.E.S.S.-era cluster survey limit. The targets are a sliding-window transient search over the full >17-year LAT exposure, any emission above ∼30 GeV flagged as non-MSP, and a 50-h CTAO-South exposure reaching $\sim 10^{-13}$ erg cm^{-2} s^{-1} at 1 TeV. The program feeds the confirmation tier of the coordination protocol and screens dark-matter-versus-exotica readings before anomaly language.

8.1 Science case

ω Cen is a GeV source: 4FGL J1326.7−4729 coincides with the core, and its emission is attributed to the MSP ensemble after deep searches found no individual pulsations (Dai et al., 2020, 2023). This is both foreground and opportunity: the steady MSP emission is a calibration source against which *transient* or *spectrally anomalous* excesses can be sought with high contrast. The technosignature channel shares the burst phenomenology of Program 5 (the Dvali and Osmanov 2023 spectrum is "democratic" across species, implying γ-ray counterparts to any neutrino burst); the conventional channel is a TeV detection or limit on the cluster, of independent interest for MSP magnetospheric and inverse-Compton physics.

8.2 Implementation

Fermi-LAT (archival, immediate). Re-analysis of the now >17-year LAT exposure at the cluster position: (i) refit of the steady source with current diffuse models; (ii) a sliding-window transient search ($\Delta t = 10^2$–10^4 s, matched to Program 5's windows) over the full mission, with the steady MSP flux as null model; (iii) spectral-cutoff analysis: MSP ensembles cut off at a few GeV, so any significant emission above ∼30 GeV flags a non-MSP component. All three are pure archival analyses.

CTAO-South (2027+). The first Paranal telescopes are, on early-2026 construction schedules, arriving through late 2026 (Cherenkov Telescope Array Consortium et al., 2019); intermediate arrays operate while construction proceeds. We propose (i) a 50-h deep exposure reaching $\sim 10^{-13}$ erg cm^{-2} s^{-1} at 1 TeV (~ 0.2 per cent of the Crab Nebula flux at 1 TeV on the HEGRA Crab spectrum, Aharonian et al. 2000) for the completed Alpha-configuration array; the 2027-era partial array delivers a factor 2–4 less, and the depth quoted here is recovered as construction completes. Either figure substantially betters the archival VHE limits placed on globular clusters by the H.E.S.S. cluster survey (H.E.S.S. Collaboration et al., 2013), making this the first CTAO-depth TeV limit on any globular cluster core. We also propose (ii) a standing target-of-opportunity trigger: any Program-5 alert within 1° pre-authorizes prompt CTAO follow-up, the latency being minutes for a source that transits Paranal $\sim 23°$ from zenith.

8.3 Decision criteria

A LAT transient coincident with a neutrino window elevates any Program-5 candidate to the campaign's highest tier. A TeV detection of steady emission is conventional astrophysics: it would make ω Cen the second globular cluster with detected VHE emission, after H.E.S.S.'s detection coincident with Terzan 5 (Abramowski et al., 2011), and the first with a resolved connection to a candidate IMBH environment. Its absence at CTAO depth tightens the inverse-Compton budget of the MSP population well beyond the existing H.E.S.S.-era cluster limits (H.E.S.S. Collaboration et al., 2013). Spectral excess above 30 GeV without transient behaviour prompts a dark-matter-versus-exotica analysis before any H_2 language is entertained.

9 Program 7: optical/IR time domain with Rubin/LSST and archival anomalies

Bottom line. The decisive observable is a nuclear tidal-disruption event, the one natural signal that would announce the central object unambiguously, together with a population-level photometric anomaly class at $\geq 3\sigma$ from the archival census. There is no current detection; the oMEGACat archive is essentially unmined for variability. The targets are an LSST broker filter flagging nuclear transients at $r \gtrsim 16$ and all-sky coverage of the saturated bright regime, plus a variability census of $\sim 10^6$ stars to ~ 0.05 mag depth over two decades of baseline. A TDE feeds the pre-arranged multi-wavelength response; anomaly classes feed the coordination protocol.

9.1 Science case

Two distinct searches share this program. First, transients: tidal disruption of a star by an IMBH is the one natural event that would announce the central object unambiguously (Hills, 1988; Gezari, 2021); rates are low ($\sim 10^{-7}$–10^{-6} yr^{-1} per cluster for full disruptions, higher for partial-disruption flares of the dense remnant environment) but the cost of monitoring is now effectively zero, because Rubin has taken science data since late 2025 (Pre-LSST), the ten-year LSST was announced as formally begun on 2026 June 30, and ω Cen sits in the survey footprint (Ivezić et al., 2019; Vera C. Rubin Observatory, 2026). Second, archival photometric anomalies: the oMEGACat HST archive (500+ epochs over 20+ years for 10^6 stars; Häberle et al. 2024a; Nitschai et al. 2023) is an essentially unmined time-domain dataset in which periodic dimmings, secular fades, or statistically anomalous variability classes (the photometric residues conjectured for megastructures; Wright et al. 2014; Hsiao et al. 2021; Socas-Navarro et al. 2021) can be sought with two decades of baseline for free.

9.2 Implementation

LSST stream and bright-transient coverage. At the cluster's distance modulus of $\mu = 13.7$, a full TDE peaking at $M_V \approx -14$ to -17 would appear at $V \approx 0$ to -3: hopelessly saturated for LSST difference imaging, and indeed for any survey-class imager. The bright-event trigger therefore belongs to ASAS-SN/Evryscope-class all-sky photometry, which monitors this magnitude range continuously; the campaign's role is to register the ω Cen position with those pipelines. Both instruments saturate well short of the TDE peak: ASAS-SN saturates at $V \approx 10$–11 mag (Kochanek et al., 2017), and Evryscope's short-exposure photometry is saturation-limited near $V \approx 6$ mag (Law et al., 2015), so the bright-transient trigger relies on the recovered, non-linear photometry of saturated frames rather than on standard aperture photometry. LSST covers the complementary faint regime: a standing broker filter on the field flags

(i) any nuclear transient in the unsaturated range $r \gtrsim 16$, which captures partial disruptions and other sub-luminous accretion flares, and (ii) anomalous variables among $\sim 10^5$ measurable members outside the saturated core. Difference-imaging in the crowded annulus is handled with the oMEGACat reference catalogue.

Archival mining. A systematic variability census of the oMEGACat photometric archive: period search to ∼0.05 mag depth across 10^6 stars, secular-trend extraction, and an explicitly pre-registered anomaly statistic (population-level excess of non-physical light-curve classes) with human vetting of survivors. Conventional yield: the deepest variable-star census of any globular cluster, eclipsing-binary distance cross-checks, and stellar-rotation demographics — publishable regardless of anomalies.

Core-depletion statistic (joint with Program 3). Paper A dependence (labelled): H_2's P4c predicts secular depletion of the loosely bound core population beyond N-body expectations. The measurement (star counts by binding-energy quantile against matched N-body models; González Prieto et al. 2025) is identical to standard mass-segregation analysis and is conventional dynamics on its face; it operationalizes Paper A's P4c observable, stated there as a mass-function-slope test, as a binding-energy-quantile star count instead. We do not pre-register the depletion direction as a primary discriminant: at the 10-pc delivery radius the Phase-4 deficit is ∼1 per cent of the enclosed mass, the removed mass reappears as central point mass, and any deficit refills on the ∼12-Gyr half-mass relaxation time (Swanson, 2026a).

9.3 Decision criteria

A nuclear TDE settles the IMBH question electromagnetically (and its decay light curve constrains $M_\bullet$; Gezari 2021); the campaign's role is to guarantee the multi-wavelength response is pre-arranged. Archival anomalies are population statistics: no single weird star means anything (the false-positive history of photometric SETI is long); only a $\geq 3\sigma$ class-level excess, surviving vetting, enters the coordination protocol.

10 Program 8: Supplementary archival channels

Bottom line. The decisive observable is an X-ray point source at the kinematic centre, at any flux; continuous-wave searches at twice the known pulsar spin frequencies and a stacked ultra-high-energy cosmic-ray directional analysis complete the messenger coverage. The current best is the ∼291 ks Chandra limit of $L_X \lesssim 1.6 \times 10^{30}$ erg s^{-1}. The targets are a re-derivation of that limit at the current centre, stacked Chandra/XMM-Newton depth, an eROSITA variability screen, O5-class CW limits of $h_0 \lesssim 10^{-26}$, and a public-data UHECR stack. The X-ray result feeds the joint fundamental-plane constraint with Programs 1 and 2.

Three channels are cheap enough to run as graduate-student archival projects and complete the messenger coverage.

Archival X-rays (Chandra, XMM-Newton, eROSITA). The X-ray band is the standard probe of quiescent black-hole accretion, and the campaign as presented so far would otherwise lack it. The benchmark limit is the ∼291 ks Chandra exposure of Haggard et al. (2013), which detects no central point source and bounds the unabsorbed luminosity at $L_X(0.5\text{–}7.0\ \mathrm{keV}) \lesssim 1.6 \times 10^{30}$ erg s^{-1}, an Eddington ratio of $\lesssim 10^{-12}$ for a $10^4\ M_\odot$ hole. The archival program re-derives this limit at the current oMEGACat kinematic centre (the 2013 analysis predates the fast-star centre determination), searching within the wander-inflated tolerance $\max(1'', r_\bullet(M_\bullet))$ of Section 5.2 rather than at a fixed position, stacks the subsequent Chandra and XMM-Newton exposures of the field, and folds in the released eROSITA all-sky survey epochs (eRASS1 in the DR1 release, plus the subsequent completed passes as the consortium

releases them) as a variability screen, noting that the survey was halted in February 2022, so the eROSITA screen covers 2020–2022 only; a joint X-ray–radio treatment through the fundamental plane (Merloni et al., 2003) then constrains radiatively inefficient flow models alongside the Program 1 and Program 2 limits. For reproducibility the re-registration is defined against stated epochs rather than against a position alone: the oMEGACat centre is propagated from its own catalogue reference epoch to the 2013 exposure's epoch using the cluster's absolute proper motion, both epochs are reported with the re-derived limit, and the search aperture is the wander-inflated $\max(1'', r_\bullet)$ rather than a point. Any X-ray point source at the kinematic centre, at any flux, would be a major result; the expected null sharpens the deepest multi-band quiescence measurement for any black hole candidate.

Continuous gravitational waves (LIGO–Virgo–KAGRA). Narrowband searches at twice the spin frequencies ($f_{\rm GW} = 2f_{\rm spin}$) of the nineteen known MSPs, using public O5 data and the MeerKAT ephemerides, following the known-pulsar methodology of Abbott et al. (2022). Expected upper limits $h_0 \lesssim 10^{-26}$ are an O5-class expectation (a factor $\gtrsim 2$ better than the best O3 targeted result, PSR J0711−6830, once converted to this frequency band) and translate, at ω Cen's own MSP population, which clusters near $P \approx 4$ ms ($f_{\rm GW} \approx 500$ Hz), to ellipticity limits $\varepsilon \approx 2.1 \times 10^{-7}$ at 5.49 kpc. Reaching a few $\times 10^{-8}$ needs a faster spinner than any known ω Cen MSP: at $P = 2.5$ ms ($f_{\rm GW} = 800$ Hz) the limit is $\varepsilon \approx 8 \times 10^{-8}$, and 3×10^{-8} requires $P \approx 1.5$ ms. This is conventional neutron-star physics in an environment (a dense old cluster) where the recycled-pulsar population is dynamically distinctive, with one complication the standard pipeline does not carry: the observed $\dot{f}$ of a cluster MSP is dominated by line-of-sight acceleration in the cluster potential rather than intrinsic spin-down, several ω Cen MSPs have measured negative $\dot{f}$ for exactly this reason, which Program 2's own timing-acceleration map exploits as signal rather than noise. The search must therefore fit $\dot{f}$ and $\ddot{f}$ as free terms rather than assume a spin-down model, and the spin-down-limit framing standard targeted searches quote is not available for these targets. The trials factor across the nineteen targets is carried in the post-trials threshold of Table 5. The technosignature reading (structured masses in MSP orbits) is not separately funded; it shares the identical data product.

Ultra-high-energy cosmic rays (Pierre Auger). ω Cen sits well inside Auger's southern acceptance. A stacked directional analysis around the cluster position in the public $E > 8$ EeV dataset (Pierre Auger Collaboration et al., 2017), with 1° and 3° apertures bracketing magnetic deflection, either contributes an upper limit on hadronic acceleration in the core (of conventional interest given the MSP wind population) or flags an anisotropy for which the Galactic-Centre region (52.4° away, but the dominant anisotropy structure at these energies) is the confounder to exclude first.

11 Coordination, joint statistics, and data policy

11.1 The two-messenger rule, operationalized

Table 5 pre-registers the campaign's trigger and confirmation matrix. Each row names a primary trigger, the channels that can confirm it, and the joint-significance requirement. Combined significances across independent channels use Fisher's method on the per-channel p-values; each per-channel p-value entering the combination must itself be post-trials within its channel (its look-elsewhere budget over windows, frequencies, and positions already paid), so the combination never launders pre-trials significances. Two opposing effects enter the combined significance; both are stated. Because post-trials p-values are conservative rather than uniform under the null, the Fisher combination inherits that conservatism. But Fisher's method assumes the inputs are independent under the null, and several of the pairings in Table 5 are not: a genuine tidal-disruption flare excites P1, P2, P7 and P8 together, and the LAT and CTAO rows share both the Galactic diffuse model and the cluster's own millisecond-pulsar population. Under positive dependence the Fisher statistic has larger variance than the χ^2_{2n} reference, so the tail probability is

understated and the method is anti-conservative. The two effects have no reason to cancel. We therefore apply Brown (1975) for any combination whose inputs are not demonstrably independent under the astrophysical null, estimating the null covariance of the per-channel statistics from archival data (which costs no observing time) and rescaling the degrees of freedom accordingly; the covariance-estimation procedure is pre-registered alongside the thresholds, though the specific archival datasets it will draw on are not yet selected and this element of the plan remains aspirational. Where the covariance is not yet estimated, the combination is restricted to channel pairs that are independent by construction, and the stated global threshold bounds rather than equals the nominal false-alarm rate. The 5×10^{-7} figure is a per-trigger-class threshold, not a campaign-wide false-alarm budget over the ten trigger rows and fifteen years of monitoring; a campaign-wide false-alarm budget aggregating all ten channels is not computed in this paper and is deferred to the individual pipeline papers. The global claim threshold is set at $p < 5 \times 10^{-7}$ post-trials — deliberately at discovery convention, because the base rate of true technosignatures is unknown and plausibly zero, so the loss function is asymmetric: a false positive damages the field's credibility far more than a delayed true positive damages the discovery (Wright et al., 2022).

This matrix is one of two pre-registrations governing the campaign, and the two occupy different layers. The Fisher-combined $p < 5 \times 10^{-7}$ matrix of Table 5 is the *trigger layer*: it decides when an event or excess is promoted from a single-channel candidate to a campaign-level anomaly requiring adjudication. Paper E's hierarchical Bayes-factor framework, with Kass–Raftery $\ln K$ action bands (Kass and Raftery, 1995) whose "candidate" band itself requires evidence in at least two messengers, is the *inference layer*: it takes any promoted trigger and adjudicates it against the full $H_q/H_{\rm sub}/H_{\rm eng}$ hypothesis set (Swanson, 2026a). The frequentist matrix therefore controls the false-alarm budget of what enters adjudication, while the Bayesian framework determines what the adjudicated evidence supports; a trigger that clears Table 5 but lands in a weak $\ln K$ band is reported as an unexplained astrophysical candidate, not an anomaly claim. One mandatory rule governs the interface between the layers. Promotion is data-dependent, so the inference layer always evaluates its likelihoods on the complete campaign record, every epoch and every channel including those that recorded nothing, never on the promoted subset alone. The trigger layer allocates follow-up effort and carries no inferential weight. Scoring only the promoted data would require an explicit selection correction of order $\ln P(\text{trigger} \mid H_0) \approx -14$ nats at the threshold above, which is large enough to move a nominal candidate to null-favoured; carrying the full record avoids the correction rather than paying it (Swanson, 2026a). The significance conventions of the two layers, and of the companion papers that consume or produce them, are set side by side in Appendix 14.

11.2 Alert latency and standing authorizations

The latency-critical path is neutrino → gamma/optical: KM3NeT real-time alerts (commissioning through 2026) distribute within seconds–minutes; the CTAO ToO and LSST-broker hooks are standing authorizations requiring no human decision for the first response hour. Slow-path coordination (JWST DDT requests, radio re-pointing) is pre-drafted as template proposals with trigger criteria attached, cutting submission latency to <24 h.

11.3 Data policy

All pipelines, thresholds, and analysis code are public from the start (the pre-registration is itself the methods paper); all derived catalogues are released with their papers; raw data inherit host-facility policies. That guarantee covers what the campaign controls: its own code, thresholds, and derived products. What host collaborations must grant is a separate matter, negotiated by memorandum of understanding and

Table 5: Pre-registered trigger and confirmation matrix. "Joint" = Fisher-combined post-trials significance required for any anomaly claim; single-channel events, however significant, are reported as astrophysical candidates only.

Primary trigger	Channel threshold	Confirming channels	Disposition
Neutrino multiplet (P5)	$\geq$ 3 tracks at 10^2–10^3 s, $\geq$ 4 at 10^4 s / 1° / $\geq$10 TeV, 5σ post-trials against the 20× background envelope, trials counted per window size	Fermi-LAT window (P6); CTAO ToO (P6); JWST/ground IR (P1)	repeat at same position required for anomaly claim; single event published as astrophysical transient
Narrowband radio line (P2)	SNR > 15; multi-beam cross-rejection; multi-epoch scintillation-aware confirmation (Section 4)	off-source cadence; second telescope (e.g. ATCA)	interference exclusion protocol precedes any claim
IR variable/transient (P1, P7)	5σ on the two-epoch difference ($\geq$ 20% amplitude above 118 nJy at NIRCam, above 5.9 μJy at F1000W; Section 3.2) or new source > 5σ	radio continuum (P2); X-ray archival; LSST stream (P7)	classified as accretion/TDE candidate first
Photometric anomaly class (P7)	$\geq 3\sigma$ population-level excess	independent archive (Gaia epochs); spectroscopy	population statistic only; no single-star claims
Astrometric anomaly (P3)	wander outside H_0/H_1 envelopes at 3σ; light-versus-heavy factor-2.4 separation only, the 17% within-light-regime split retired on degrees of freedom (Section 5.2); gated on a measured N-body model-to-model scatter below 2.7 μas yr^{-1}, on verified Roman crowded-field astrometric performance, and on the F146 $\approx$ 16–21 reference window (core-depletion statistic retired as a primary discriminant: at the 10-pc delivery radius the Phase-4 deficit is ~1 per cent of the enclosed mass, the removed mass reappears as central point mass, and any deficit refills on the ~12-Gyr half-mass relaxation time; Harris 1996; Swanson 2026a)	timing acceleration map (P2)	feeds LISA prior; no standalone claim
>30 GeV spectral excess (P6)	$\geq 5\sigma$ non-MSP component in LAT refit	CTAO follow-up (P6); radio/X-ray positional coincidence (P2, P8)	dark-matter-versus-exotica analysis precedes any anomaly language
CW detection at MSP frequency (P8)	$\geq 5\sigma$ in known-pulsar narrowband search	EM timing solution and position (P2)	classified as neutron-star physics first
UHECR anisotropy (P8)	$\geq 3\sigma$ post-trials stacked excess, 1°–3° apertures	Fermi-LAT/CTAO coincidence (P6); Galactic-Centre confounder excluded	directional statistic only; no source claim without γ-ray support
X-ray point source at centre (P8)	any source > 3σ at kinematic centre	radio continuum (P2); JWST IR counterpart (P1)	classified as accretion candidate; feeds fundamental-plane fit
LISA spin/mass result (P4)	per mission parameter estimation	all of the above as context	adjudicates $H_0/H_1/H_2$ per Section 13

Table 6: Campaign summary. "Type": A = archival/commensal, N = new observations, F = free by-product of funded survey. Costs are rough full-cost estimates (Section 12); facility time is listed separately since it is allocated, not purchased.

#	Program	Type	Key sensitivity	Facility request	Timeline	Decides
P1	JWST IR limits	N	~10 nJy (F444W); $L_{\rm bol} \lesssim 10^{30}$ erg s^{-1}	~65 h, 2 epochs	2027–2029	accretion state; waste heat
P2	Radio: timing + SETI + continuum	N	$\sigma_a \sim 10^{-10}$ m s^{-2} (5 yr, white-noise extrapolation); EIRP 4×10^{15} W (incoh., full cluster)	~130 h yr^{-1} MeerKAT, 5 yr	2026–2031+ → SKA-Mid	H_0 vs H_1 (profile); first SETI limit
P3	Astrometry: DR4 + Roman + ELT	A/F/N	frame to DR4; wander factor-2.4 (light vs heavy); inner-star discovery	archival; Roman survey; ~28 h ELT (dedicated)	2026–2035	mass tension; wander mass
P4	LISA forecast	F	$\delta M/M \sim 10^{-3\text{–}4}$, $\delta a_\star \sim 10^{-3}$ (if inspiral)	none (mission science)	2035+	everything (Tier 3)
P5	Neutrino monitoring	A/N	$E^2\Phi \sim$ few$\times 10^{-12}$ TeV cm^{-2} s^{-1}; burst triplets	pipeline on ARCA stream	2026–2040	Dvali–Osmanov channel
P6	Gamma rays	A/N	LAT transient windows; 10^{-13} erg cm^{-2} s^{-1} at 1 TeV (50 h)	50 h CTAO + ToO	2026–2032	burst counterparts; first CTAO-depth TeV limit
P7	Time domain	A/F	TDE flares unmissable; 10^6-star archival census	LSST stream; archival	2026–2036	TDE; anomaly classes
P8	X-ray + CW + UHECR archival	A	$L_X \lesssim 10^{30}$ erg s^{-1}; $h_0 \lesssim 10^{-26}$; stacked UHECR limit	public data	2026–2029	completeness

costed in Table 7: MeerKAT and SKA raw data fall under SARAO proprietary policy, and KM3NeT does not release track-level data outside the collaboration, so the Program-5 burst pipeline runs either inside the collaboration on the full stream or externally on the much smaller public alert stream. Negative results are published on a fixed, pre-announced cadence, so that the decision to publish never depends on what the data showed.

11.4 Program summary

Table 6 assembles the eight programs with sensitivities, requests, and timelines.

12 Cost and timeline

12.1 Costing

Table 7 presents full-cost estimates (salaries, computing, travel, publication; US full-cost academic rates including overheads) for each program. Facility time is excluded, because it is competitively allocated and carries no cash cost to the campaign, as is LISA, which is mission science.

That exclusion is what the "marginal cost" framing means, and it should not be read as a small facility request. The dedicated component is ~130 h yr^{-1} of MeerKAT over five years (~650 h of bi-weekly pointings), 65 h of JWST across two epochs, 50 h of CTAO, and ~28 h of ELT. Commensal or archival are the SETI spectrometer backend (which rides the timing sessions), the Gaia DR4 re-derivation, the Chandra, XMM-Newton and eROSITA stacks, the LSST and Roman survey epochs, and the ARCA and LAT analyses. Three cash items the table prices are consequences of the dedicated hours rather than of the archival work: the data-management line above, the host-collaboration MoUs, and the SETI backend's compute. The KM3NeT track-level access line ($120k over ten years) buys agreement and coordination, and it does not buy an FTE inside that collaboration; it assumes in-kind analysis effort from a collaboration member, and the campaign states that dependency rather than costing around it. Two further lines deserve

Table 7: Indicative full-cost budget (US$k, 2026 dollars). Personnel figures assume postdoc-led programs with PI/Co-I fractions and graduate students where noted. Estimates for the Roman component of P3 and for P5, P6, P7 are new to this paper; others follow the per-instrument proposal studies available at https://omegacentauri.me.

Program	Cost	Years	Dominant element
P1 JWST deep imaging + waste heat	1,310	3	postdoc + grad + HPC
P1b JWST spectroscopic phase (contingent)	257	2	postdoc (triggered only)
P2 MeerKAT timing + SETI + continuum	2,900	5	2 postdocs + grad + computing
P3 HST/Gaia frame + Roman wander	560	4	postdoc + HPC (full-cost rates)
P3b ELT/MICADO epochs	490	4	postdoc share + travel
P4 LISA preparatory modelling	150	3	PI/Co-I fractions
P5 Neutrino burst pipeline	380	3	postdoc + compute (full-cost rates)
P6 Fermi archival + CTAO ToO	200	3	postdoc share
P7 LSST broker + archival mining	300	3	postdoc + broker engineering
P8 X-ray + CW + UHECR archival	239	3	graduate RA + HPC
Data management (P2 commensal product)	450	5	∼90 TB yr^{-1} archive + transport at SARAO
Host-collaboration MoUs (SARAO, KM3NeT)	120	10	data-access agreements
Coordination, data releases, workshops	150	10	part-time coordinator
Subtotal (P1b excluded)	7,249	—	—
Contingency (20%)	1,450	—	—
Total (P1b excluded)	**8,699**	—	—

comment. The data-management line prices the Program-2 commensal product, which is not free, and the quantity it prices needs its averaging stated. A 1-Hz channelization of MeerKAT's 856-MHz L band in two polarizations is 1.7×10^9 spectral samples per dump, so raw 1-s dumps at 32-bit precision would be 24.7 TB hr^{-1} and even 8-bit 1-s dumps 6.2 TB hr^{-1}. The ∼0.7 TB hr^{-1} figure corresponds to 8-bit samples averaged to ∼8-s dumps (0.77 TB hr^{-1}), which is the adopted configuration; quoted without it the number implies 0.91 bits per spectral sample. At that configuration the reduced product is ∼90 TB yr^{-1} at the requested cadence. Transient voltage buffers are a separate line and are not folded into the 0.7 TB hr^{-1}; archiving and transport are budgeted at SARAO. The drift-search sensitivity of Eq. (1) is set by the 1-Hz channel width rather than by the dump time, so the averaging costs the SETI analysis nothing except time resolution on signals varying faster than 8 s. The MoU line covers the host-collaboration data-access agreements of Section 11. With a 20 per cent contingency the total is ≈ US$8.7M ($7,249k plus 20 per cent) across the 2026–2040 span the programs actually occupy, rather than the 2026–2035 window quoted through Draft v2.6: Program 5 runs to 2040 in Figure 5 and the host-collaboration MoUs are ten-year agreements. It is still small by every relevant comparison: a single mid-scale NASA Explorer instrument, one JWST cycle's archival-funding pool, or the marginal cost of a few nights of ELT operations.

12.2 Timeline

Figure 4 lays the programs against the facility schedule. The structure is deliberate: the 2026–2028 window is dominated by archival work and standing-pipeline construction (cheap, immediate); 2029–2034 by the new-facility harvest (ELT, SKA-Mid, CTAO completion, Roman mid-mission); 2035+ by LISA. Decision points D1–D4 are marked and defined in Section 13.

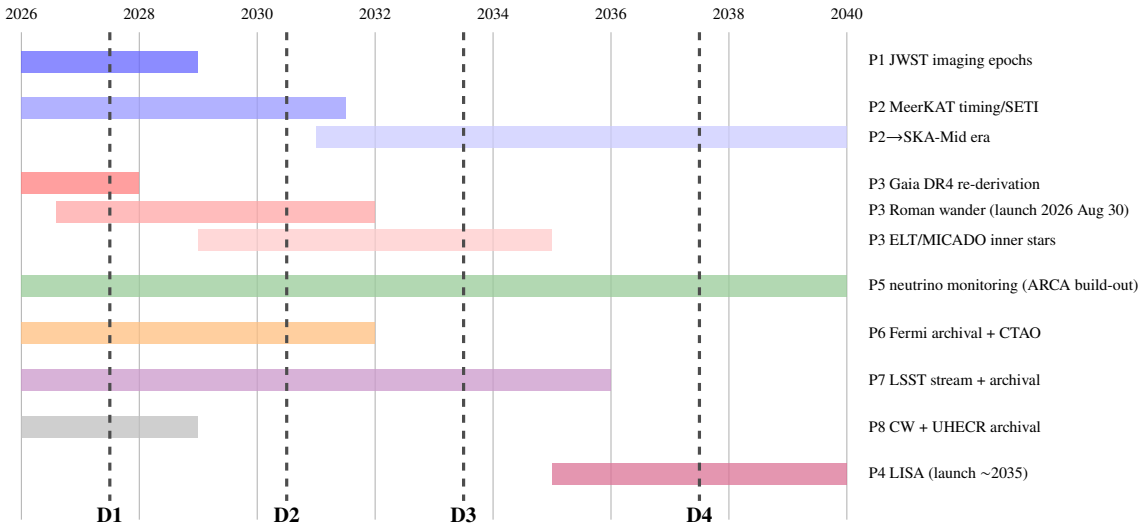


Figure 4: Campaign timeline against the facility schedule, 2026–2040. Decision points: D1 (2027): Gaia-DR4 re-derivation of the fast-star bound. D2 (2030): five-year pulsar acceleration profile; Roman wander first results. D3 (2033): pre-LISA synthesis — H_0 versus H_1 called at available significance. D4 (2037): LISA verdict, if a source is caught.

13 The decision structure through 2040

Figure 5 lays the facility schedule, the program phases, and the D1–D4 decision points on a single time axis; Figure 6 then assembles the campaign into a single decision tree, extending the gravitational-wave branch of Paper A's falsification framework backward into the electromagnetic-and-timing era. Rough branch weights are stated where the inputs exist to estimate them; they are planning priors, not forecasts, and we expect D1/D2 to revise them substantially. The tree's light-branch mass cutoff and spin boundary are taken verbatim from Paper A's falsification matrix (Table 7 of Swanson 2026b, tests T4 and T5) ; Paper A is the fence owner and no fence is restated independently here. The tree reproduces Paper A's version 2.11 gravitational-wave tree: a mass node at the $8.2 \times 10^3\ M_\odot$ kinematic lower bound that demotes the selection ranking (T4, a demotion and not a kill), then a spin split at $a_\star = 0.1$ (T5 kill) and $a_\star = 0.9$, with 0.1–0.9 grey. For the spin prior we adopt Paper A's: ω Cen is the nucleus of an accreted dwarf, coherent disc growth generically reaches high spin, and that spin survives the Blandford–Znajek spindown time, so a measured $a_\star \gtrsim 0.9$ is expected on H_0 as well and is not discriminating on spin alone (Swanson, 2026b). Program 4's isotropic-capture estimate, $a_\star = 0.049 \pm 0.021$ at the endpoint mass to 0.084 at the fiducial mass, is the alternative channel; it falls inside the T5 kill region.

Several numbers in this paper are imported from companions that are preprints and are not yet refereed: from Paper E the Bayes-factor action bands, the waste-heat floor, the regulation statistic R, the Phase-4 deficit and its refill clock, the isotropic-capture spin estimate and its $\epsilon_{\rm rot}$ tolerance; from Paper A the T4 and T5 fences and the spin prior; from Paper H the mass-tension framing, which itself rests on the refereed Häberle et al. 2024b bound. We mark thresholds that depend on those companions provisional, and a reader who declines them should treat the corresponding thresholds as stated without numerical values.

13.1 Amendment C-A1 (2026-08-26, pre-data)

An external audit round identified three defects in the decision tree as printed through Draft v2.3: the D4 mass partition left $[5 \times 10^3, 10^4)\ M_\odot$ firing no branch at all, an interval that contains the fast-star kinematic

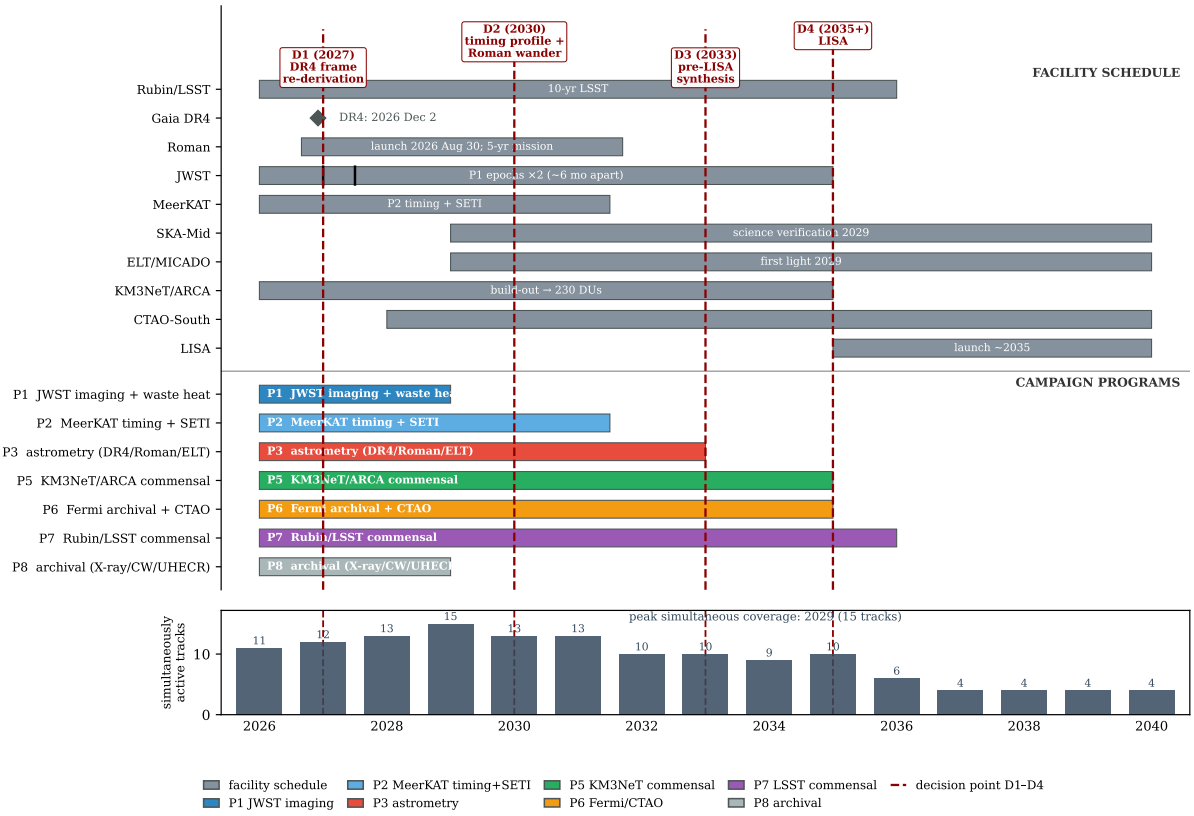


Figure 5: Campaign timeline as a Gantt chart, 2026–2040. Top block: the facility schedule (grey), with Gaia DR4 as a diamond marker at 2026 December 2, the two P1 JWST epochs as ticks on the JWST bar, the KM3NeT/ARCA bar ending at the 230-detection-unit build-out, and the LISA bar starting at the ~2035 launch. Middle block: the observing programs P1–P8 (Table 6), coloured as in Figure 4. Vertical dashed lines: the decision points D1 (2027, Gaia-DR4 frame re-derivation), D2 (2030, five-year timing profile and Roman wander), D3 (2033, pre-LISA synthesis), and D4 (2035+, LISA; Section 13). Bottom strip: the number of simultaneously active facility and program tracks per year. Coverage peaks in 2029, when SKA-Mid and ELT/MICADO come online while the MeerKAT, JWST, and KM3NeT programs are still running; the decisive measurements cluster at D1–D3 and at the LISA window.

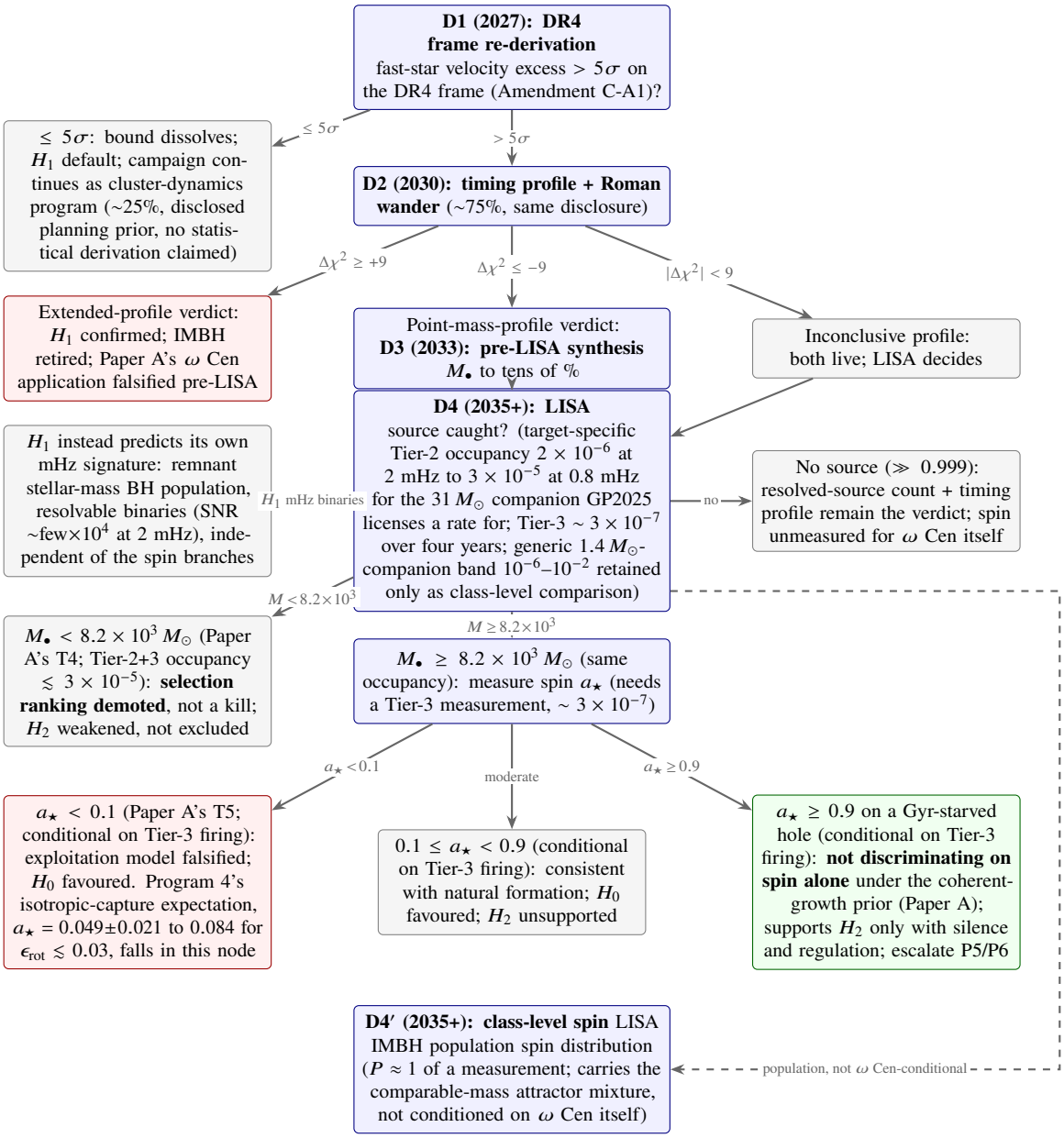


Figure 6: Decision structure through 2040. Red boxes retire hypotheses (for the $a_\star < 0.1$ node, the exploitation model); the single green branch is the only path on which the technosignature interpretation gains material support, and even there the claim is "anomaly requiring explanation." The green branch is the thin-disc spin-up branch (Paper E, Section 3.4): on a MAD-mode spin-up the endpoint is $a_\star \approx 0.035$–0.07, at or below the natural-capture null and below the T5 trigger $a_\star < 0.1$, i.e. indistinguishable from the null, and an engineered hole on that branch would fall in the red $a_\star < 0.1$ node. The mass split carries the Tier-2+3 occupancy and the spin split the Tier-3 probability. Percentages and weights are disclosed planning priors rather than computed probabilities, stated at the target-specific rate where GP2025 licenses one; D1's split is binary because any excess at or below the 5σ contradiction threshold reverts to the same systematics-bias reading, not because a graded middle state was considered and discarded. The colours alone can mislead in two places. First, the mass fence is the 8,200 $M_\odot$ velocity-only lower bound itself, so a measurement below it demotes the selection ranking and contradicts the fast-star analysis without falsifying H_2, consistent with the statement in Section 15 that no null result in this campaign carries evidential weight against H_2. Second, the wander and neutrino rows feeding D2 and D3 are systematics-gated rather than decision-grade until their gates are met (the N-body scatter floor of Section 5.2 and the window-dependent multiplicity of Section 7); where a gate fails, the branch carries no weight and D2 routes on the timing statistic alone. What revises D1's 25/75 weights is stated in advance: a DR4 frame shift that moves the fast-star velocities below the escape speed pushes weight toward the systematics reading, and one that leaves the excess intact at $> 5\sigma$ pushes it the other way, so the prior is revised by the size and sign of the frame correction rather than by a judgement call after the fact. D2's routing follows Amendments C-A1 and C-A2 below, sourced to the executed mock forecast (`paper/figs/fC_calc7b_d2forecast_v2.py`). The routing statistic is the fixed-parameter one, twice the

lower bound of 8,200 $M_\odot$; the D2 routing into the extended, point-mass, and inconclusive branches carried only qualitative labels while the upstream mock-forecast gate was already numeric; and D1's binary split disclosed no basis for its 25/75 weights. All three are pre-registered-rule defects, so none is repaired silently, following the disclosure convention of the mass-tension companion's Amendments A1–A5. No observation has yet been scored against the tree: every program in this paper remains pre-D1, and this amendment changes only the tree's own text, not any result.

The repair follows a ratified panel review (`board/notes/R7-PANEL-DECIDE.md`, brief B-1, option (a)) of three ledger findings (`C-M5`, `C-M6`, `C-t13`). First, the D4 gap is closed by splitting the light branch at Paper A's own T4 fence ($5 \times 10^3\ M_\odot$, adopted verbatim, not restated independently): masses below it retire H_2 as before, and the newly covered $[5 \times 10^3, 10^4)\ M_\odot$ interval is scored as a weakened, not falsified, selection ranking, exactly as Paper A's own T4 language already treats that mass range (a $5 \times 10^3\ M_\odot$ hole still delivers $\sim 6 \times 10^{34}$ W). Second, the D2 routing adopts the numeric thresholds $\Delta\chi^2_{\rm fixed} \geq +9$ (extended), ≤ -9 (point mass), and $|\Delta\chi^2_{\rm fixed}| < 9$ (inconclusive), taken from this paper's own pre-registered mock-forecast criterion (Section 4, `0xAlpha/results/R7-CALC-C2D.md`). Amendment C-A2 below revises what the subscript means and bounds where the rule applies. Third, D1's 25/75 split is left in place but is now stated explicitly as a disclosed planning prior with no claimed statistical derivation, matching this figure's own long-standing caption convention; the binary structure is retained because any excess that fails to survive the DR4 re-derivation at $> 5\sigma$ collapses to the same systematics-bias reading regardless of degree, not because a graded middle outcome was considered and discarded.

Superseded in part by Amendment C-A3 below (the T4 fence value and kill/demotion status).

13.2 Amendment C-A2 (2026-09-03, pre-data)

Amendment C-A1 named a numeric D2 routing threshold without naming which of the forecast's statistics it applies to, and justified the threshold with an equivalence that holds for only one of them. Re-running the forecast on a corrected joint likelihood (Section 4) shows the ambiguity is not cosmetic: the two statistics disagree on the same data, and the free-parameter one is uninformative. This amendment fixes the routing statistic, bounds the rule's scope, and records the four defects found in the machinery. As with C-A1, no observation has been scored against the tree; every program remains pre-D1.

The routing statistic is the fixed-parameter one. D2 routes on $\Delta\chi^2_{\rm fixed}$, twice the log-likelihood ratio between the two *named* potentials with no free parameters, signed positive toward the extended component. The equivalent evidence band is $\ln K \geq +4.5$ (extended) or ≤ -4.5 (point mass), and for this statistic alone $\Delta\chi^2 = 2 \ln K$ exactly. The free-parameter statistic, which profiles each family over its own grid, is reported as a diagnostic and routes nothing. Read on the free-parameter statistic, the C-A1 threshold of ±9 would return an inconclusive verdict in nearly every realization at every noise floor tested, since its median under an injected $2 \times 10^4\ M_\odot$ point mass is at most +0.02, and the sign of that statistic is wrong in 52–79 per cent of realizations.

Scope. The rule adjudicates the two named potentials and nothing else. On the mass-matched third truth of Section 4 it routes a genuinely extended component to the point-mass branch in 98.5 and 80 per cent of realizations at the two lowest floors and returns an inconclusive verdict in 92.5–100 per cent once the noise floor reaches 31.6 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc}^{-1}$, so a D2 verdict of "point mass" is a verdict against the named $2.5 \times 10^5\ M_\odot$ extended model and not against extended models in general. Where the mass-matched cell fails, D2 weight shifts to the astrometric channels of Section 5 on the same rule C-A1 already states for a short forecast.

Machinery defects, disclosed rather than repaired silently. The v2 script corrects four faults in the executed forecast, and the v1 script is retained so the chain stays auditable. The family fits summed each pulsar's own marginal likelihood over the mass grid, which let every pulsar choose an independent central

mass instead of testing one common potential; both the profile maximum and the evidence are now formed on the joint likelihood across pulsars. A unit constant converting $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$ to $\mathrm{m\,s^{-2}}$ was wrong by a factor 10^{22} and disagreed with its own comment; the correct value is 3.2408×10^{-11}. Validation gate $\mathrm{G_A}$ was reported as failing and the failure was not surfaced, and the gate was itself mis-constructed: it tested a $2 \times 10^4\ M_\odot$ mass at $|r| = 0.42$ pc including the visible component, against a printed number for a $4 \times 10^4\ M_\odot$ point mass at 0.3 pc. With both the constant and the construction repaired the gate reproduces the printed $6 \times 10^{-8}\ \mathrm{m\,s^{-2}}$ to 3 per cent, and $\mathrm{G_B}$, previously one-sided and unfailable, now records that the printed $1\text{–}4 \times 10^{-9}\ \mathrm{m\,s^{-2}}$ brackets the median of the extended envelope while the envelope itself spans 3.9 decades. The cell results are unaffected by the unit error, since the noise ladder is dimensionless in the same units; the gates were not. Before the fourth fault was found, a provenance gate confirmed the fixed-parameter statistic was unchanged from v1 to 1.3×10^{-14}, so every difference between the two runs was attributable to the likelihood repair. The fourth fault was found on 2026-10-22: every dark-mass term in the cell path omitted the gravitational constant, so the injected and fitted dark signal was $1/G = 232.5$ times too large against the visible acceleration and the noise ladder. The gates did not catch it because they call the machinery's own acceleration routine, which carries G, and never touch the cell path. The cells quoted in Section 4 and in this amendment were regenerated with G applied (Draft v3.4); a new gate $\mathrm{G_D}$ checks the cell-path dark acceleration against GMz/r^3 to 10^{-12} and against the machinery's own routine, and the v1 provenance check, which held between two runs that shared the omission, is now informational.

13.3 Amendment C-A3 (2026-10-20, pre-data)

Amendment C-A1 split the light branch of D4 at $5 \times 10^3\ M_\odot$, drew the branch below it as a kill, and described that fence as adopted verbatim from Paper A. Paper A version 2.11 sets its T4 fence at $8.2 \times 10^3\ M_\odot$ (the Häberle et al. 2024b velocity-only lower bound), calls T4 a demotion of the selection ranking and not a kill, and adds a T5 kill at $a_\star < 0.1$ with $0.1 \le a_\star < 0.9$ grey. The tree as printed through Draft v3.2 therefore differed from the document whose fences it claimed to copy, and it had no node for the $a_\star < 0.1$ region that contains Program 4's own isotropic-capture expectation ($a_\star = 0.049 \pm 0.021$ to 0.084). This changes pre-registered branch rules, so it is disclosed here and C-A1 is left as written.

The tree of Figure 6 is rebuilt from Paper A version 2.11: a mass node at $8.2 \times 10^3\ M_\odot$ that demotes the selection ranking and does not retire H_2; above it a spin split at $a_\star = 0.1$ (kill of the exploitation model, T5) and $a_\star = 0.9$ (green, not discriminating on spin alone under Paper A's coherent-growth prior), with 0.1–0.9 grey. The spin prior is Paper A's, and the earlier "$\to 0$ under H_0's null" weight on the green branch is withdrawn. The mass split is labelled with the Tier-2+3 occupancy ($\lesssim 3 \times 10^{-5}$) and only the spin split with the Tier-3 probability ($\sim 3 \times 10^{-7}$). The isotropy tolerance quoted on the green branch is $\epsilon_{\mathrm{rot}} \approx 0.03$ (Paper E: 0.026–0.031). No observation has been scored against the tree; every program remains pre-D1.

13.4 Amendment C-A4 (2026-10-22, pre-data)

The criterion of $\Delta\chi^2_{\mathrm{fixed}} \ge 9$ and the ±9 routing thresholds are unchanged. This amendment designates 31.6 to 64 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$ as the D2-epoch noise, on the evidence that the formal timing error is negligible and intrinsic spin-down dominates, and records that the criterion is missed there for both named truths. Under C-A1 and Section 4, D2 weight therefore rests on the astrometric channels. Timing profiles resume routing authority only if a constraint on intrinsic spin-down reaching 10 is demonstrated before the D2 data lock. No observation has been scored.

14 Significance conventions across the companion papers

The campaign and its companion papers score evidence in four conventions, and a number quoted in one vocabulary must not be silently read in another. This appendix states the four side by side.

Post-trials p-values (this paper). The frequentist false-alarm probability of a trigger after the look-elsewhere correction over the search's windows, frequencies, and positions. The action boundary is $p < 5 \times 10^{-7}$ per trigger class (Table 5), the discovery convention, with cross-channel combinations handled as in Section 11. It applies to the trigger layer of all eight programs in this paper.

$\ln K$ **Bayes factors (Paper E).** The logarithm of the ratio of evidences between two hypotheses, taken by Paper E as $H_{\rm eng}$ against the best astrophysical null. The action bands follow Kass and Raftery (1995) as pre-registered in Paper E: below 0 null-favoured, 0 to 1 uninformative, 1 to 3 anomaly, 3 to 5 strong anomaly, and above 5, sustained across at least two messengers, candidate. The look-elsewhere treatment is structural: likelihoods are always evaluated on the complete campaign record, never on the promoted subset alone, so the selection effect is priced rather than corrected after the fact, and prior sensitivity is reported as a band rather than a point. It applies to Paper E's adjudication layer, acting on whatever this paper's trigger layer promotes.

95 per cent credible upper limits (Paper F). The Bayesian one-sided credible bound on a parameter, in Paper F the Bondi efficiency $\epsilon_{\rm B}$ or the dark mass. There is no action band; the limit is the product rather than a decision. The corresponding treatment is prior sensitivity, reported alongside the bound, since a credible limit conditions on the model instead of testing it. It applies to Paper F's quiescent-accretion limits on ω Cen and on the comparison clusters.

Posterior mass fractions (Paper H). The fraction of posterior probability in a region of the dark-component mass and scale-radius plane under one normalized model. There is no discovery threshold; the pre-registered criteria are consistency checks, sign stability of the Bayes factor across every prior cell and posterior-predictive adequacy in each cell. The prior-grid sweeps are themselves the sensitivity treatment, and a verdict is quotable only when it survives the full pre-registered cell set. It applies to Paper H's joint fast-star and pulsar fit.

No single convention is privileged; the choice depends on whether the question is "should we follow up?" (frequentist), "what does the evidence support?" (Bayes factor), "what is the bound?" (credible limit), or "is the model consistent?" (posterior check).

15 Conclusion

Omega Centauri in 2026 presents a configuration that observational astronomy rarely supplies: a contested discovery (the Galaxy's best IMBH candidate) whose two strongest constraints formally contradict each other; an electromagnetic silence deep enough to be a result in itself; an essentially untouched technosignature parameter space; and a fleet of new southern and all-sky instruments (Rubin, Roman, Gaia DR4, KM3NeT, CTAO, SKA-Mid, ELT, and ultimately LISA) arriving on the timescale needed to resolve all of it. The campaign presented here is designed to use that conjunction fully.

Its architecture reflects three commitments. First, *dual use*: every program stands as conventional astrophysics (the mass tension, the quiescent-accretion frontier, the MSP dynamical laboratory, the first deep TeV and neutrino limits on a globular cluster), so that no null result is a loss. Second, *conservative*

sensitivity accounting: where the obvious measurement fails (direct fast-star acceleration before ∼2040), we demonstrate the failure quantitatively and redesign around it, because a program that oversells its tests invites the credibility collapse that has repeatedly damaged this field. Third, *pre-registration*: hypotheses $H_0/H_1/H_2$, thresholds, confirmation rules, and decision points are fixed in advance, so that whatever the sky delivers (a remnant swarm, a quiet heavy hole, or a genuine anomaly), the inference chain was written down before anyone knew the answer.

The adjudication asymmetry deserves a blunt statement. Under realistic outcomes, this campaign will resolve H_0 versus H_1; it will neither support nor retire H_2 except through the low-probability LISA spin branch or a serendipitous multi-messenger multiplet. No null result carries evidential weight against H_2, because the leakage-free and electromagnetically silent predictions of that hypothesis class are compatible with every null the campaign can produce; what the campaign buys for H_2 is the pre-registered infrastructure to adjudicate a positive trigger if one ever arrives, and quantitative limits on the specific loud sub-classes (beacons, warm waste heat above the JWST floor[2], neutrino bursts at predicted strength).[3]

The cost is US$8.7M across 2026–2040 including 20% contingency (Table 7), most of it salaries for archival analysis riding on facilities that are funded regardless. The payoff structure is asymmetric in the campaign's favour: the probable outcomes (a resolved mass tension, a characterized central object, the deepest technosignature limits on 10^7 old stars) are solid science at modest cost, and the improbable outcome (the green branch of Figure 6) would justify the program many thousands of times over. We commit, symmetrically, to the red branches: if the timing profile confirms an extended remnant component, or LISA delivers a slowly spinning ($a_\star < 0.1$) hole, the exploitation model retires here without special pleading (a light hole demotes ω Cen's selection ranking without retiring H_2), and what remains is what was always underneath: the most interesting stellar system in the southern sky, finally measured properly.

16 Revision record

This appendix lists what changed between versions where a reader may have quoted an earlier number or fence.

Draft v3.5 (2026-10-22). Citation-audit corrections. The isotropic-capture spin expectation of Program 4 is now 0.049 ± 0.021 at the endpoint mass and 0.084 at the fiducial mass (Paper E), replacing $\approx 0.06 \pm 0.02$ in the Program 4 discussion, the decision tree and Amendment C-A1; both values remain inside the T5 region $a_\star < 0.1$. The Harris-catalogue tidal radius of 1.06° is replaced by $r_c\ 10^c \approx 0.81°$ in the narrowband-SETI beam argument. The Paper A references are updated to version 2.11 and to its falsification-matrix table number, and the LSST start of 2026 June 30 now cites the Rubin Observatory announcement. The Merritt et al. (2007) page range in the bibliography is corrected to 553–563. No forecast value, fence or branch rule changes.

Draft v3.4 (2026-10-22). The D2 forecast cells of Section 4 were regenerated after a units bug: the cell path omitted the gravitational constant on every dark-mass term (Amendment C-A2, machinery defects). Values withdrawn: the fixed-parameter medians −93 to −419 (compact truth) and $+1.1 \times 10^4$ to $+4.7 \times 10^7$ (extended truth) are now −0.27 to −23 and +0.20 to +436; the free-parameter median +79 is now +0.02, with correct-sign recovery 21–48 per cent (was 11–18); the mass-matched cells (misroute 91–100 per cent at 31.6 and above) are now routed to the point-mass branch in 98.5, 80 and 7.5 per cent at the three lowest

[2]Waste heat need not be warm: computation cooled toward the 2.7 K cosmic microwave background floor would radiate grey, line-free thermal emission in the far infrared and submillimetre, with the required radiating area growing as T^{-4}, a regime the mid-infrared bands of Program 1 do not reach (Garrett, 2026).

[3]The inventory of channels limited here is deliberately small; the wider space of proposed technosignatures, from gravitational-wave beacons to the by-products of interstellar transport, is catalogued by Jackson and Benford (2020).

floors and inconclusive above; and the claim that the fixed-parameter statistic routes correctly in every realization of every compact-truth cell no longer holds (99.5, 84, 9.5, 0 and 0 per cent across the ladder). The consequence for the timing channel is recorded in Amendment C-A4: the D2 criterion is met at 3.09 and 10 $(\mathrm{km\,s^{-1}})^2\,\mathrm{pc^{-1}}$ and missed at 31.6 and above, and unknown intrinsic spin-down sets the effective noise at 31.6 to 64. The Program 2 white-noise figure of 10^{-10} $\mathrm{m\,s^{-2}}$, the Program 2 bottom line and the timing paragraph of the falsification section were made conditional on that noise condition.

Draft v3.3 (2026-10-20). (i) The Program 2 narrowband-SETI five-year allocation is 390–650 h (26 sessions per year of 3–5 h), not the 156–260 h two-year total; at the $t^{-1/2}$ scaling of Eq. (1) the threshold falls by a factor 2.4–3.0 to ~ 1.3–1.7×10^{15} W, not to 2–3 $\times 10^{15}$ W. (ii) The decision tree of Figure 6 is rebuilt on Paper A version 2.9 (Amendment C-A3): the mass node sits at $8.2 \times 10^3\,M_\odot$ (a demotion node, replacing the $5 \times 10^3\,M_\odot$ kill box and the $[5 \times 10^3, 10^4)\,M_\odot$ sub-floor node), an $a_\star < 0.1$ node (T5) is added, and the green branch no longer goes to zero under the H_0 null but is non-discriminating on spin alone under Paper A's coherent-growth prior. The mass split carries the Tier-2+3 occupancy and the spin split the Tier-3 probability. (iii) The isotropy tolerance $\epsilon_{\rm rot}$ is ≈ 0.03 (Paper E: 0.026–0.031), not 0.035. (iv) The variability band of Paper E's statistic R is 10^{-2}–10^{-1} Hz, matching cycle times of 10–100 s, not 10^{-3}–10^{-1} Hz. (v) The Phase-4 refill clock is the ~ 12 Gyr half-mass relaxation time, not the ~ 4 Gyr core time; the retirement of the core-depletion statistic stands on its other two grounds. (vi) The clause comparing the 21,100 $M_\odot$ acceleration-informed bound with the timing limit was removed from the mass-tension discussion; the argument rests on 8,200 > 6,000 $M_\odot$. (vii) The earlier contrast of "$\simeq 0.7$ versus $\gtrsim 0.9$" for the Tier-3 spin reading was removed from Program 4.

Draft v3.2 and earlier. Amendment C-A1 (2026-08-26) and Amendment C-A2 (2026-09-03) remain in Section 13 as the pre-data record of the D4 mass partition and the D2 routing statistic. Draft v2.6 values withdrawn in the body are: the two-epoch variability trigger "≥20–30 per cent for any source above ∼30 nJy" (Program 1), the 9.8 effective-baseline figure and the "$\Delta\chi^2$ of order tens" forecast expectation (Program 2), the $\sigma_a \approx 1 \times 10^{-3}$ mas yr^{-2} astrometric figures, the uniform-cadence curve and the "3–5σ discrimination" claim (Program 3), the $\sim 5 \times 10^{-7}$ ($\approx 5\sigma$) neutrino-window significance (Program 6), and the 2026–2035 span quoted for the programme timeline and cost.

Acknowledgements and disclosure

The author thanks the maintainers of the NASA Astrophysics Data System and arXiv, on which the citation verification for this work relied. **AI assistance disclosure:** drafting, citation verification, derivation checking, and figure preparation for this manuscript were performed with substantial assistance from a large language model (Claude, Anthropic), under the author's direction; the author reviewed and takes full responsibility for all claims, derivations, and references. Interactive calculators implementing the quantitative material in this paper, together with the underlying per-instrument proposal studies, are available at https://omegacentauri.me.

Data availability

No new observational data were generated for this work. All quantitative claims derive from the cited literature; analysis conventions for the proposed programs will be released with the respective pipeline papers. The pre-registration documents (hypotheses, trigger matrix, thresholds, and analysis plans, as fixed by Tables 6 and 5) are maintained at https://omegacentauri.me.

References

Aartsen, M. G. et al. (2020). Time-integrated neutrino source searches with 10 years of IceCube data. *Physical Review Letters*, 124:051103.

Abbott, R., Abe, H., Acernese, F., Ackley, K., Adhikari, N., Adhikari, R. X., et al. (2022). Searches for gravitational waves from known pulsars at two harmonics in the second and third LIGO-Virgo observing runs. *The Astrophysical Journal*, 935:1.

Abramowski, A. et al. (2011). Very-high-energy gamma-ray emission from the direction of the galactic globular cluster terzan 5. *Astronomy & Astrophysics*, 531:L18.

Adrián-Martínez, S. et al. (2016). Letter of intent for KM3NeT 2.0. *Journal of Physics G: Nuclear and Particle Physics*, 43(8):084001.

Aharonian, F. A. et al. (2000). The energy spectrum of TeV gamma rays from the Crab Nebula as measured by the HEGRA system of imaging air Cerenkov telescopes. *The Astrophysical Journal*, 539:317–324.

Albert, A., André, M., Anghinolfi, M., Ardid, M., Aubert, J.-J., et al. (2020). ANTARES and IceCube combined search for neutrino point-like and extended sources in the Southern Sky. *The Astrophysical Journal*, 892:92.

Álvarez-Domínguez, Á., Garay, L. J., Martín-Martínez, E., and Polo-Gómez, J. (2024). No black holes from light. *Physical Review Letters*, 133:041401.

Amaro-Seoane, P. (2018). Relativistic dynamics and extreme mass ratio inspirals. *Living Reviews in Relativity*, 21:4.

Amaro-Seoane, P. et al. (2017). Laser interferometer space antenna. LISA L3 mission proposal submitted to ESA.

Anderson, J. and van der Marel, R. P. (2010). New limits on an intermediate-mass black hole in Omega Centauri. I. Hubble Space Telescope photometry and proper motions. *The Astrophysical Journal*, 710(2):1032–1062.

Arca Sedda, M., Amaro-Seoane, P., and Chen, X. (2021). Merging stellar and intermediate-mass black holes in dense clusters: Implications for LISA, LIGO, and the next generation of gravitational wave detectors. *Astronomy & Astrophysics*, 652:A54.

Aros, F. I. and Vesperini, E. (2023). Effects of massive central objects on the degree of energy equipartition of globular clusters. *Monthly Notices of the Royal Astronomical Society*, 525:3136–3148.

Babak, S., Gair, J., Sesana, A., Barausse, E., Sopuerta, C. F., Berry, C. P. L., Berti, E., Amaro-Seoane, P., Petiteau, A., and Klein, A. (2017). Science with the space-based interferometer LISA. V. Extreme mass-ratio inspirals. *Physical Review D*, 95(10):103012.

Bañares-Hernández, A., Calore, F., Martín Camalich, J., and Read, J. I. (2025). New constraints on the central mass contents of Omega Centauri from combined stellar kinematics and pulsar timing. *Astronomy & Astrophysics*, 693:A104.

Baumgardt, H. and Vasiliev, E. (2021). Accurate distances to Galactic globular clusters through a combination of Gaia EDR3, HST, and literature data. *Monthly Notices of the Royal Astronomical Society*, 505:5957–5977.

Bhathal, R. (2010). The australian optical SETI project. In *Astrobiology Science Conference 2010*. Abstract 5046.

Braun, J., Dumm, J., De Palma, F., Finley, C., Karle, A., and Montaruli, T. (2008). Methods for point source analysis in high energy neutrino telescopes. *Astroparticle Physics*, 29:299–305.

Braun, R., Bonaldi, A., Bourke, T., Keane, E., and Wagg, J. (2019). Anticipated performance of the Square Kilometre Array – Phase 1 (SKA1). *arXiv e-prints*.

Breen, P. G. and Heggie, D. C. (2013). Dynamical evolution of black hole subsystems in idealized star clusters. *Monthly Notices of the Royal Astronomical Society*, 432:2779–2797.

Brown, M. B. (1975). A method for combining non-independent, one-sided tests of significance. *Biometrics*, 31:987–992.

Chatterjee, P., Hernquist, L., and Loeb, A. (2002). Dynamics of a massive black hole at the center of a dense stellar system. *The Astrophysical Journal*, 572:371–381.

Chen, S., Hare, J., Kargaltsev, O., Yang, H., Cioffi, D., Häberle, M., and Seth, A. (2026). The intermediate mass black hole in Omega Centauri: Constraints on accretion from JWST. *The Astrophysical Journal*.

Chen, W., Freire, P. C. C., Ridolfi, A., Barr, E. D., Stappers, B., Kramer, M., et al. (2023). MeerKAT discovery of 13 new pulsars in Omega Centauri. *Monthly Notices of the Royal Astronomical Society*, 520:3847–3856.

Chen, X., Vázquez-Aceves, V., Chen, S., Lee, K., Guo, Y., and Liu, K. (2025). Detecting intermediate-mass black holes using miniature pulsar timing arrays in globular clusters. *arXiv e-prints*, page arXiv:2507.08201.

Cherenkov Telescope Array Consortium, Acharya, B. S., Agudo, I., Al Samarai, I., et al. (2019). *Science with the Cherenkov Telescope Array*. World Scientific, Singapore.

Clontz, C., Seth, A. C., Dotter, A., Häberle, M., Nitschai, M. S., Neumayer, N., Feldmeier-Krause, A., Latour, M., Wang, Z., Souza, S. O., Kacharov, N., Bellini, A., Libralato, M., Pechetti, R., van de Ven, G., and Alfaro-Cuello, M. (2024). oMEGACat. IV. Constraining the ages of Omega Centauri subgiant branch stars with HST and MUSE. *The Astrophysical Journal*, 977:14.

Cohen, N., Malkan, M., and Dickey, J. (1980). A passive SETI in globular clusters at the hydroxyl and water lines. *Icarus*, 41:198–204.

Colom i Bernadich, M., Dai, S., Abbate, F., Kerr, M., Bachetti, M., Bhargava, Y., Buchner, S., Johnston, S., Burgay, M., Possenti, A., Nag, R., Ridolfi, A., Carleo, A., Corongiu, A., Freire, P. C. C., Camilo, F., Chen, W., Cadelano, M., Risbud, D., Padmanabh, P. V., Champion, D. J., Kramer, M., Stappers, B., Serylak, M., Balakrishnan, V., Bailes, M., Dutta, A., Vleeschower Calas, L., Venkatraman Krishnan, V., and Men, Y. (2026). A joint MeerKAT and Parkes view of Omega Centauri: New TRAPUM searches and pulsar timing. *arXiv e-prints*, page arXiv:2603.21845.

Colpi, M. et al. (2024). LISA definition study report. arXiv:2402.07571; ESA-SCI-DIR-RP-002.

Condon, J. J. (1974). Confusion and flux-density error distributions. *The Astrophysical Journal*, 188:279.

Condon, J. J. (2007). Deep radio surveys. In Afonso, J., Ferguson, H. C., Mobasher, B., and Norris, R., editors, *At the Edge of the Universe: Latest Results from the Deepest Astronomical Surveys*, volume 380 of *ASP Conference Series*, page 189.

Condon, J. J., Cotton, W. D., Greisen, E. W., Yin, Q. F., Perley, R. A., Taylor, G. B., and Broderick, J. J. (1998). The nrao vla sky survey. *The Astronomical Journal*, 115(5):1693–1716.

Cordes, J. M., Lazio, T. J. W., and Sagan, C. (1997). Scintillation-induced intermittency in seti. *The Astrophysical Journal*, 487:782–808.

Crawford, I. A. (2026). Some thoughts on the future of technosignature searches: Constraining the Fermi paradox. To appear in Proceedings of IAU Symposium 404.

Curtis, O., Adams, V. H., Angerhausen, D., Bates, J., Berea, A., Dick, S. J., Elvis, M., Khatri, S. P., Linares, R., Muco, M., Seager, S., and Wright, J. T. (2026). The Dyson Minds 2025 workshop: SETI around black holes. *Publications of the Astronomical Society of the Pacific*, 138:046001.

Czech, D., Isaacson, H., Pearce, L., Lebofsky, M., Price, D. C., MacMahon, D. H. E., et al. (2021). The Breakthrough Listen search for intelligent life: MeerKAT target selection. *Publications of the Astronomical Society of the Pacific*, 133:064502.

Dai, S. et al. (2020). Discovery of millisecond pulsars in the globular cluster Omega Centauri. *The Astrophysical Journal Letters*, 888:L18.

Dai, S., Johnston, S., Kerr, M., Berteaud, J., Bhattacharyya, B., Camilo, F., and Keane, E. (2023). Timing of pulsars in the globular cluster Omega Centauri. *Monthly Notices of the Royal Astronomical Society*, 521:2616–2622.

Davies, R. et al. (2021). MICADO: The multi-adaptive optics camera for deep observations. *The Messenger*, 182:17–21.

Dvali, G. and Osmanov, Z. N. (2023). Black holes as tools for quantum computing by advanced extraterrestrial civilizations. *International Journal of Astrobiology*, 22(6):617–640.

Dyson, F. J. (1960). Search for artificial stellar sources of infrared radiation. *Science*, 131(3414):1667–1668.

Enriquez, J. E., Siemion, A., Foster, G., Gajjar, V., Hellbourg, G., Hickish, J., Isaacson, H., Price, D. C., Croft, S., DeBoer, D., et al. (2017). The Breakthrough Listen search for intelligent life: 1.1–1.9 GHz observations of 692 nearby stars. *The Astrophysical Journal*, 849:104.

Event Horizon Telescope Collaboration (2022). First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way. *The Astrophysical Journal Letters*, 930:L12.

Fragione, G., Leigh, N. W. C., Ginsburg, I., and Kocsis, B. (2018). Tidal disruption events and gravitational waves from intermediate-mass black holes in evolving globular clusters across space and time. *The Astrophysical Journal*, 867:119.

Freire, P. C. C., Ridolfi, A., Kramer, M., Jordan, C., Manchester, R. N., Torne, P., et al. (2017). Long-term observations of the pulsars in 47 Tucanae – II. Proper motions, accelerations and jerks. *Monthly Notices of the Royal Astronomical Society*, 471:857–876.

Gaia Collaboration, Prusti, T., de Bruijne, J. H. J., Brown, A. G. A., Vallenari, A., Babusiaux, C., et al. (2016). The Gaia mission. *Astronomy & Astrophysics*, 595:A1.

Garrett, M. (2026). Slysh haloes: the waste heat of cold computing as a submillimetre technosignature.

Gezari, S. (2021). Tidal disruption events. *Annual Review of Astronomy and Astrophysics*, 59:21–58.

González Prieto, E., Rodriguez, C. L., and Cabrera, T. (2025). Growing the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal Letters*, 990:L69.

GRAVITY Collaboration, Abuter, R., Amorim, A., Anugu, N., Bauböck, M., Benisty, M., et al. (2018). Detection of the gravitational redshift in the orbit of the star S2 near the Galactic centre massive black hole. *Astronomy & Astrophysics*, 615:L15.

Greene, J. E., Strader, J., and Ho, L. C. (2020). Intermediate-mass black holes. *Annual Review of Astronomy and Astrophysics*, 58:257–312.

Häberle, M., Neumayer, N., Bellini, A., Libralato, M., Clontz, C., Seth, A. C., et al. (2024a). oMEGACat. II. Photometry and proper motions for 1.4 million stars in Omega Centauri and its rotation in the plane of the sky. *The Astrophysical Journal*, 970:192.

Häberle, M., Neumayer, N., Clontz, C., Seth, A. C., Bellini, A., et al. (2025). oMEGACat. VI. Analysis of the overall kinematics of Omega Centauri in 3D: Velocity dispersion, kinematic distance, anisotropy, and energy equipartition. *The Astrophysical Journal*, 983:95.

Häberle, M., Neumayer, N., Seth, A., Bellini, A., Libralato, M., Baumgardt, H., Whitaker, M., Dumont, A., Alfaro-Cuello, M., Anderson, J., Clontz, C., et al. (2024b). Fast-moving stars around an intermediate-mass black hole in ω Centauri. *Nature*, 631(8020):285–288.

Haggard, D., Cool, A. M., Heinke, C. O., van der Marel, R., Cohn, H. N., Lugger, P. M., and Anderson, J. (2013). A deep Chandra X-ray limit on the putative IMBH in Omega Centauri. *The Astrophysical Journal Letters*, 773:L31.

Harris, W. E. (1996). A catalog of parameters for globular clusters in the Milky Way. *The Astronomical Journal*, 112:1487.

H.E.S.S. Collaboration, Abramowski, A., et al. (2013). Search for very-high-energy gamma-ray emission from Galactic globular clusters with H.E.S.S. *Astronomy & Astrophysics*, 551:A26.

Hilker, M. and Richtler, T. (2000). ω Centauri – a former nucleus of a dissolved dwarf galaxy? New evidence from Strömgren photometry. *Astronomy & Astrophysics*, 362:895–909.

Hills, J. G. (1988). Hyper-velocity and tidal stars from binaries disrupted by a massive Galactic black hole. *Nature*, 331:687–689.

Hsiao, T. Y.-Y., Goto, T., Hashimoto, T., Santos, D. J. D., On, A. Y. L., Kilerci-Eser, E., Wong, Y. H. V., Kim, S. J., Wu, C. K.-W., Ho, S. C.-C., and Lu, T.-Y. (2021). A dyson sphere around a black hole. *Monthly Notices of the Royal Astronomical Society*, 506(2):1723–1732.

Huang, B.-L., Tao, Z.-Z., and Zhang, T.-J. (2026a). Phase-space crystallization in Galactic globular clusters: A Gaia-based metric and implications for technosignature searches. *The Astrophysical Journal*, 1005:102.

Huang, B.-L., Tao, Z.-Z., Zhang, T.-J., and Gajjar, V. (2026b). The FAST-SETI Milky Way globular cluster survey. I. A pilot multibeam on-the-fly search of five globular clusters at the L band. *The Astronomical Journal*, 171(1):51.

Ibata, R. A., Bellazzini, M., Malhan, K., Martin, N., and Bianchini, P. (2019). Identification of the long stellar stream of the prototypical massive globular cluster ω Centauri. *Nature Astronomy*, 3:667–672.

Ivezić, Ž., Kahn, S. M., Tyson, J. A., Abel, B., Acosta, E., Allsman, R., et al. (2019). LSST: From science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873:111.

Jackson, A. A. and Benford, G. (2020). Novel technosignatures.

Jonas, J. L. and MeerKAT Team (2016). The MeerKAT radio telescope. In *Proceedings of MeerKAT Science: On the Pathway to the SKA*, volume MeerKAT2016 of *Proceedings of Science*, page 001.

JWST GO 8322 team (2026). Astrometric monitoring of the fast stars around the IMBH in ω Centauri. JWST Proposal, GO 8322. https://www.stsci.edu/jwst-program-info/download/jwst/pdf/8322/.

Kass, R. E. and Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795.

Kochanek, C. S., Shappee, B. J., Stanek, K. Z., Holoien, T. W.-S., Thompson, T. A., Prieto, J. L., Dong, S., Shields, J. V., Will, D., Britt, C., Perzanowski, D., and Pojmański, G. (2017). The all-sky automated survey for supernovae (asas-sn) light curve server v1.0. *Publications of the Astronomical Society of the Pacific*, 129(980):104502.

Lacki, B. C. and DiKerby, S. (2025). Possibilities for SETI at high energy. arXiv:2506.16351; white paper submitted to the 2025 NASA DARES RFI.

Lacki, B. C. et al. (2021). One of everything: The Breakthrough Listen exotica catalog. *The Astrophysical Journal Supplement Series*, 257:42.

Law, N. M., Fors, O., Ratzloff, J., Wulfken, P., Kavanaugh, D., Sitar, D. J., Pruett, Z., Birchard, M. N., Barlow, B. N., Cannon, K., Cenko, S. B., Dunlap, B., Kraus, A., and Maccarone, T. J. (2015). Evryscope science: Exploring the potential of all-sky gigapixel-scale telescopes. *Publications of the Astronomical Society of the Pacific*, 127(949):234.

Loeb, A. (2024). Comment on "No black holes from light". arXiv:2408.06714; reply by original authors at arXiv:2408.11097.

Mahida, A. D. et al. (2026). No evidence for accretion around the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal*, 996:122.

Mandel, I., Brown, D. A., Gair, J. R., and Miller, M. C. (2008). Rates and characteristics of intermediate mass ratio inspirals detectable by Advanced LIGO. *The Astrophysical Journal*, 681:1431–1447.

Martinez, M. A. S., González Prieto, E., and Rasio, F. A. (2026). Survival analysis of intermediate-mass black holes in dense star clusters. *arXiv e-prints*, page arXiv:2602.23431. Submitted to ApJ.

Merloni, A., Heinz, S., and Di Matteo, T. (2003). A fundamental plane of black hole activity. *Monthly Notices of the Royal Astronomical Society*, 345:1057–1076.

Merritt, D., Berczik, P., and Laun, F. (2007). Brownian motion of black holes in dense nuclei. *The Astronomical Journal*, 133:553–563.

Nitschai, M. S., Neumayer, N., Clontz, C., Häberle, M., Seth, A. C., Husser, T.-O., et al. (2023). oMEGACat. I. MUSE spectroscopy of 300,000 stars within the half-light radius of ω Centauri. *The Astrophysical Journal*, 958:8.

Noyola, E., Gebhardt, K., and Bergmann, M. (2008). Gemini and Hubble Space Telescope evidence for an intermediate-mass black hole in ω Centauri. *The Astrophysical Journal*, 676:1008–1015.

Pierre Auger Collaboration, Aab, A., Abreu, P., Aglietta, M., et al. (2017). Observation of a large-scale anisotropy in the arrival directions of cosmic rays above 8×10^{18} eV. *Science*, 357:1266–1270.

Prager, B. J., Ransom, S. M., Freire, P. C. C., Hessels, J. W. T., Stairs, I. H., Arras, P., and Cadelano, M. (2017). Using long-term millisecond pulsar timing to obtain physical characteristics of the bulge globular cluster Terzan 5. *The Astrophysical Journal*, 845:148.

Robson, T., Cornish, N. J., and Liu, C. (2019). The construction and use of LISA sensitivity curves. *Classical and Quantum Gravity*, 36:105011.

Seth, A. C. et al. (2024). Weighing the intermediate mass black hole in Omega Centauri. JWST Proposal, Cycle 3, GO 5137. https://www.stsci.edu/jwst/phase2-public/5137.pdf.

Socas-Navarro, H., Haqq-Misra, J., Wright, J. T., Kopparapu, R., Benford, J., Davis, R., and TechnoClimes 2020 workshop participants (2021). Concepts for future missions to search for technosignatures. *Acta Astronautica*, 182:446–453.

Soltis, J., Casertano, S., and Riess, A. G. (2021). The parallax of ω Centauri measured from Gaia EDR3 and a direct, geometric calibration of the tip of the red giant branch and the Hubble constant. *The Astrophysical Journal Letters*, 908:L5.

Souza, S., Neumayer, N., Seth, A. C., Wang, Z., Clontz, C., Häberle, M., Nitschai, M. S., Smith, P. J., Matsuno, T., et al. (2026). oMEGACat. X. shedding light on the disrupted dwarf galaxy of Omega Centauri. *arXiv e-prints*, page arXiv:2603.23589. Submitted to ApJ.

Stappers, B. and Kramer, M. (2016). An update on TRAPUM. In *Proceedings of MeerKAT Science: On the Pathway to the SKA*, volume MeerKAT2016 of *Proceedings of Science*, page 009.

Strader, J., Chomiuk, L., Maccarone, T. J., Miller-Jones, J. C. A., Seth, A. C., Heinke, C. O., and Sivakoff, G. R. (2012). No evidence for intermediate-mass black holes in globular clusters: Strong constraints from the JVLA. *The Astrophysical Journal Letters*, 750:L27.

Swanson, T. (2026a). Engineered intermediate-mass black hole systems: Infrastructure constraints, observable residue, and a multi-messenger adjudication framework. Preprint; omegacentauri.me.

Swanson, T. (2026b). The macro transcension hypothesis: Spinning black holes in dense stellar clusters as thermodynamic attractors for advanced civilizations, with Omega Centauri as an observational test bed. Preprint; omegacentauri.me/paper.html.

Swanson, T. (2026c). A radio and x-ray bound on bondi-fed accretion onto the candidate intermediate-mass black hole in omega centauri, with an infrared consistency check. Preprint; omegacentauri.me.

Tarter, J. (2001). The search for extraterrestrial intelligence (SETI). *Annual Review of Astronomy and Astrophysics*, 39:511–548.

The KM3NeT Collaboration (2025). Observation of an ultra-high-energy cosmic neutrino with KM3NeT. *Nature*, 638(8050):376–382.

Tremou, E., Strader, J., Chomiuk, L., Shishkovsky, L., Maccarone, T. J., Miller-Jones, J. C. A., et al. (2018). The MAVERIC survey: Still no evidence for accreting intermediate-mass black holes in globular clusters. *The Astrophysical Journal*, 862:16.

van der Marel, R. P. and Anderson, J. (2010). New limits on an intermediate-mass black hole in Omega Centauri. II. Dynamical models. *The Astrophysical Journal*, 710:1063–1088.

van Haasteren, R. and Levin, Y. (2013). Understanding and analysing time-correlated stochastic signals in pulsar timing. *Monthly Notices of the Royal Astronomical Society*, 428:1147–1159.

Vera C. Rubin Observatory (2026). Action! NSF–DOE Vera C. Rubin Observatory begins capturing the greatest cosmic movie ever made. Press release, 2026 June 30. Accessed 2026-10-22.

Vernstrom, T., Scott, D., Wall, J. V., Condon, J. J., Cotton, W. D., Fomalont, E. B., Kellermann, K. I., Miller, N., and Perley, R. A. (2014). Deep 3 ghz number counts from a p(d) fluctuation analysis. *Monthly Notices of the Royal Astronomical Society*, 440(3):2791–2809.

WFIRST Astrometry Working Group, Sanderson, R. E., Bellini, A., Casertano, S., Lu, J. R., Melchior, P., et al. (2019). Astrometry with the Wide-Field Infrared Space Telescope. *Journal of Astronomical Telescopes, Instruments, and Systems*, 5:044005.

Whitaker, M., Kerr, E., Seth, A., Häberle, M., Strader, J., Anderson, J., Bellini, A., Clontz, C., Freeman, Z., Griggio, M., Kamann, S., Libralato, M., Neumayer, N., González Prieto, E., Rodriguez, C. L., Saracino, S., Smith, P., van de Ven, G., and Wang, Z. (2026). A long period stellar-mass black hole binary in ω Centauri. *The Astrophysical Journal Letters*.

Wright, J. T., Haqq-Misra, J., Frank, A., Kopparapu, R., Lingam, M., and Sheikh, S. Z. (2022). The case for technosignatures: Why they may be abundant, long-lived, highly detectable, and unambiguous. *The Astrophysical Journal Letters*, 927(2):L30.

Wright, J. T., Mullan, B., Sigurdsson, S., and Povich, M. S. (2014). The ĝ infrared search for extraterrestrial civilizations with large energy supplies. I. Background and justification. *The Astrophysical Journal*, 792(1):26.

Zocchi, A., Gieles, M., and Hénault-Brunet, V. (2019). The effect of stellar-mass black holes on the central kinematics of ω Cen: a cautionary tale for IMBH interpretations. *Monthly Notices of the Royal Astronomical Society*, 482:4713–4725.